

Human Capital, Competition and Mobility in the Managerial Labor Market*

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Abstract

We pose a structural model of the managerial labor market with general and firm-specific human capital accumulation, managerial bargaining power, and imperfect labor market competition. Empirically, firm-specific skill drives wage growth and variability over careers; it also restricts mobility and helps explain the low rate of external CEO hiring. We decouple bargaining power from labor market competition in managerial rent extraction, with competition accounting for a substantial share, especially for poached CEOs. Firm-specific skill accumulation increases match productivity between firms and managers, shaping the dynamics of rent extraction by raising rent growth over tenure while dampening it over experience.

Keywords: managerial human capital, CEOs, the market for CEOs, executive pay, executive mobility, on-the-job search, structural estimation, human capital accumulation, firm-specific human capital

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1. Introduction

Understanding the nature of managerial human capital, particularly the relative importance of general and firm-specific skill, is fundamental for understanding the managerial labor market. Prevailing theories, such as [Gabaix and Landier \(2008\)](#) and [Murphy and Zabojnik \(2007\)](#), attribute rising CEO pay to competition for general managerial skills. Implicit in these models is a competitive labor market with minimal frictions to mobility. However, empirical evidence has shown that external CEO hires are rare, CEO mobility is low, and the CEO labor market has become more rigid with time ([Cziraki and Jenter, 2022](#); [Barry et al., 2025](#)). These findings suggest that firm-specific skill may play a larger role in the managerial labor market than previously considered.¹

We address this apparent disconnect by posing and estimating a dynamic model of a frictional managerial labor market, featuring general and firm-specific human capital accumulation, imperfect competition for managerial talent, bargaining between firms and managers over compensation, and both internal and external promotions to CEO. This framework allows us to quantify the relative importance of the components of human capital, and separately quantify the effects of bargaining power from competitive forces in driving managerial pay.

We provide three main findings. First, firm-specific human capital plays an important role in the managerial labor market. Firm-specific skill increases the match-specific productivity between the firm and manager, and we show that its accumulation accounts for roughly 30% (50%) of compensation growth across a manager’s labor market experience (tenure at a firm). But, firm-specific skill is also lost during job transitions, raising the cost of external hires as poaching firms must compensate this loss to incentivize cross-firm transitions. This mobility friction has important implications for observed CEO hiring: firm-specific human capital raises the proportion of internal CEO hires (relative to external) by about 8 percentage points (from 66% to 74%).²

Second, we empirically decouple pure managerial bargaining power from labor market competition in determining managerial rent extraction. We find that roughly 47% of realized man-

¹[Cziraki and Jenter \(2022\)](#) show that 73% of new CEOs at S&P500 firms are internal hires from 1992-2012. [Barry et al. \(2025\)](#), using a longer sample from 1920-2023, show that CEO mobility peaked around 2000 and has fallen since.

²As discussed below, the remaining disparity is driven by search frictions and the lower cost of internal hires.

agerial rents stem from labor market competition, and that the effects of competition compound over tenure and labor market experience. Further, externally-hired CEOs benefit significantly more from competition than their internally-hired counterparts. When a poaching firm pursues an outside CEO, it must outbid the incumbent for the manager’s services; hiring grants are endogenously bid up in proportion to the productivity of the incumbent (Fee and Hadlock, 2003) and any firm-specific human capital lost in the move. By contrast, internal promotions are negotiated largely in the absence of such direct competition. By quantifying the importance of these channels, our paper provides insight into the sources of CEO rent extraction, which is essential for understanding frictions in the pay-setting process (Edmans et al., 2017).

Third, we show that firm-specific human capital has a subtle effect on the dynamics of rent extraction over a manager’s career. Because firm-specific human capital raises the match-specific productivity between firm and manager, it changes the relative bargaining incentives of incumbent and poaching firms. As firm-specific skill accumulates, it redirects rent extraction from external transitions (managers gaining surplus by taking attractive outside offers) to internal renegotiations (incumbents ceding surplus by matching attractive outside offers). Empirically, we find that firm-specific human capital roughly doubles the growth of managerial surplus capture over tenure at a particular firm. Our paper can thus give a market-based explanation for low managerial mobility and high realized rent extraction.

While most of the executive labor market literature emphasizes the firm’s demand for executive talent, we pose our model from the perspective of the manager, studying how job opportunities, human capital, and managerial bargaining power shape compensation and mobility over careers. We extend the equilibrium job search model of Bagger et al. (2014). Managers may be employed as an executive (non-CEO), or as the CEO of the firm.³ Managers accumulate human capital while employed and occasionally receive competing job offers. Building upon Bagger et al. (2014), which features general human capital accumulation, we introduce two novelties to match the market for managers. First, we allow for both firm-specific and general human capital. Sec-

³To limit confusion, throughout the paper we refer to non-CEO executives as “executives.”

ond, we incorporate both internal and external CEO promotion opportunities, and importantly, allow the arrival rates of internal and external offers to differ, which allows us to disentangle the effects of firm-specific skill from a potential preference for insiders (He and Schroth, 2024) on observed managerial mobility.

We model managerial compensation as a piece-rate contract: managers receive an endogenous portion $R \in [0, 1]$ of their contribution to firm output, following an important strand of the labor search literature (Postel-Vinay and Robin, 2002; Cahuc et al., 2006; Bagger et al., 2014). While this approach abstracts from specific features of the contract (e.g., equity awards versus cash bonuses), it allows for closed-form equilibrium contracts and provides a tractable way to characterize how bargaining power, labor market competition, and human capital separately influence surplus capture. When a manager receives an attractive outside offer, it triggers a bargaining game between the manager, incumbent, and poacher (in the spirit of Rubinstein, 1982), the outcome of which determines whether the manager accepts the outside offer or remains with the incumbent at a potentially renegotiated contract. On-the-job search can thus lead to stochastic, discrete increases in pay, even if the manager ultimately stays in their current position.

A key novelty of our model is that it provides a full characterization of how firm-specific human capital shapes labor market outcomes. Beyond inducing job lock, firm-specific skills also strengthen the incumbent firm’s incentive to match attractive outside offers, due to the rising match-specific productivity of managers over tenure. We show theoretically how firm-specific human capital affects equilibrium compensation contracts and mobility patterns, speaking to the broader literature on human capital accumulation and matching in labor markets (Jovanovic, 1979; Felli and Harris, 1996; Moscarini, 2005; Bagger et al., 2014). Upon estimating the model, we show that firm-specific human capital plays a quantitatively important role in shaping the dynamics of managerial compensation.

We estimate the model (via indirect inference) on a panel of managerial careers spanning 1992–2023, combined with hand-collected data on managerial tenure and labor market experience. The sample contains information on the dynamics of compensation, promotions, and lateral career

moves. The observed rate of managerial mobility is quite low: less than 1% of managers move laterally across firms in a given year, in line with prior literature (Parrino, 1997; Cziraki and Jenter, 2022; Barry et al., 2025).

Our estimation produces several results. First, we quantify the drivers of managerial compensation by decomposing wage growth into mobility (moving to more productive firms), labor market competition (contractual improvements from job offers), and human capital (general and firm-specific). Firm-specific (general) human capital accounts for roughly 30% (25%) of compensation growth over labor market experience and 50% (40%) of growth over tenure within a firm, making firm-specific skill a central determinant of pay. Human capital accumulation also explains about 15% of dispersion in managerial compensation, and its covariance with unobservables such as labor market competition is economically large. These patterns clarify important biases in reduced-form wage models for executives, where estimated returns to experience and tenure confound human capital accumulation with shifts in bargaining outcomes and competition.

We also analyze how firm-specific human capital impacts firms’ propensity to hire internal vs. external CEOs. By simulating a counterfactual without firm-specific human capital, we find that the internal hire share falls by 8 percentage points (from 74% to 66%, a 12% decrease). Relative to a 50/50 “hiring parity” benchmark, this change is quite large. The remaining disparity is driven by search frictions (internal promotion opportunities arrive more frequently than external) and cross-firm competition (incumbents willing to match a share of outside offers).

Our second set of results concerns realized managerial surplus capture and labor market competition. Our estimation allows us to decompose rent extraction into pure bargaining power and a labor market competition effects (as inspired by Cahuc et al., 2006). We find that, on average, managers capture roughly 49% of rents, closely in line with estimates from papers in the literature (e.g., Taylor, 2013). Our estimates, however, indicate that inherent managerial bargaining power is modest: in the absence of competition, managers would capture only about 26% of rents: roughly 47% of the surplus that managers ultimately capture is attributable to competition.⁴ Further, we

⁴As we discuss in Section 6, on average managers realized rent capture is 49%, so the portion attributable to labor market competition is $1 - 26\%/49\% \approx 47\%$.

find that CEOs’ realized surplus varies depending on if the CEO was internally promoted or externally poached, with poached CEOs commanding the greatest share of rents: internal (poached) CEOs capture 50% (80%) of total surplus, with 47% (67%) of this surplus capture driven by labor market competition. These results show that endogenizing labor market competition is crucial for understanding rent extraction in executive compensation ([Bebchuk and Fried, 2003](#)).

Lastly, we show that firm-specific human capital shapes realized CEO surplus capture. We study how CEOs’ shares of rents evolve over both tenure and labor market experience in both the baseline estimation and in a counterfactual without firm-specific human capital. Over experience, firm-specific human capital accumulation lowers the manager’s piece-rate growth (job lock reduces the contractual gains from job hopping). Over tenure, however, firm-specific human capital increases piece-rate growth as the firm is willing to match attractive offers to retain accumulated firm-specific skill. These results arise from our model’s theoretical predictions concerning firm-specific skill and managers’ realized surplus capture, with a large share of these patterns driven by a selection effect: as firm-specific skill accumulates and match quality improves, managers facing the same sequence of job offers are endogenously selected into staying at their current firms, amplifying observed surplus capture for those matches.

While our framework captures important features of the managerial labor market, we abstract away from the information and incentive frictions emphasized in prior work. For example, we do not allow the incumbent firm to have private information about managerial match quality or firm-specific human capital present in models with asymmetric employer learning in internal versus external hiring (e.g., [Waldman and Zabojnik, 2025](#)).⁵ We also do not micro-found the details of incentive pay in the presence of agency frictions such as moral hazard (e.g., [Lyman, 2024](#)).

Our approach complements these literatures by taking a different angle: rather than focusing on optimal contract design in bilateral firm–manager relationships under information and agency frictions, we study equilibrium outcomes in a competitive managerial labor market, with special

⁵Our assumption of perfect information is mainly technical: introducing asymmetric information and endogenous private beliefs about match quality into our dynamic, multi-firm bargaining environment would add substantial complexity and preclude closed-form equilibrium contracts ([Postel-Vinay and Robin, 2002](#)). Solving such a model would instead require high-dimensional numerical methods, as in [Adenbaum et al. \(2024\)](#).

emphasis on human capital accumulation and its interaction with labor market imperfections. This allows us to analyze the dynamic forces shaping managerial pay, mobility, and internal versus external promotion decisions in a rich, market-level environment.

Literature Review. This paper contributes to several strands of literature. First, we engage with the literature on managerial human capital and CEO compensation, notably the debate on the relative importance of general versus firm-specific managerial skills. [Murphy and Zabojnik \(2004, 2007\)](#) model the firm’s choice of an internal vs. an external candidate, and show that, in a market with an elastic supply of managerial labor, greater importance of general skill can rationalize the observed increase in executive pay. [Custódio et al. \(2013\)](#) provide an empirical proxy for general managerial skill and show that it is associated with higher total CEO pay, supporting [Murphy and Zabojnik \(2007\)](#). Via our structural estimation, we provide the first direct empirical quantification of the relative importance of general and firm-specific skills.

Second, our paper contributes to the literature on managerial labor market dynamics and mobility ([Cziraki and Jenter, 2022](#); [Barry et al., 2025](#)), by quantifying how firm-specific human capital restricts managers’ movement across firms. Prior literature identifies information frictions and preferences for insiders as central to explaining low managerial mobility: [Taylor \(2010\)](#) finds that managerial entrenchment can help explain low CEO turnover. [He and Schroth \(2024\)](#) find that strong preferences for internal candidates are needed to explain the large share of internal relative to external CEO hires. [Capron et al. \(2024\)](#) find similar results and show further that boards’ superior ability to screen internal candidates gives insiders an edge over outside competitors.

Importantly, these papers restrict attention to firm decision-making, treating managers as passive objects in the firm’s optimization problem. We complement these papers by integrating both firm and manager decision-making. Managers actively choose among competing offers, accumulate human capital, and bargain over pay. Firms negotiate over pay contracts in the face of external competition and decide which managers to retain. We explicitly analyze managers’ option to accept external promotions or reject in favor of a future internal promotion offer, allowing us to study the role of firm-specific skill accumulation on managers’ upward career trajectories.

Several papers show that hiring patterns at smaller firms reflect a greater reliance on general CEO skill. For example, [Kecht et al. \(2025\)](#) document that private and smaller public firms increasingly hire CEOs with externally acquired experience, consistent with a stronger emphasis on general managerial human capital. Together with this literature, our findings point to a divergence across the firm size distribution in the types of managerial skills firms require.

Third, we contribute to the literature on managerial rent extraction and bargaining power ([Bebchuk and Fried, 2003](#); [Taylor, 2013](#); [Coles et al., 2014](#)). Though existing studies recognize the separate roles of bargaining power and labor market competition in CEO compensation, they are often empirically indistinguishable. Following [Cahuc et al. \(2006\)](#), we use our model to separate these two components and quantify their contributions to CEO rent extraction. In the context of corporate executives, the high observed share of managerial rents is often taken as evidence of managerial influence on the pay-setting process. Our results highlight that a sizable portion of these rents arise endogenously from labor market competition.

Our paper relates to the literature on organizational capital, which emphasizes that intangible, firm-specific assets are often embodied in key talent. [Eisfeldt and Papanikolaou \(2013\)](#) show that this gives key employees a claim on cash flows and [Lustig et al. \(2011\)](#) show that managerial compensation helps share this risk between managers and shareholders. Our model provides a specific mechanism for how firms bargain over such firm-specific human capital and how this process affects labor market outcomes.

Our paper also contributes to the labor economics literature, particularly regarding human capital accumulation, job search and mobility. Canonical papers have typically emphasized the role of general human capital and on-the-job search on wage dynamics ([Flinn, 2006](#); [Bagger et al., 2014](#)). We integrate general and firm-specific human capital into an equilibrium framework and study how firm-specific human capital acts as both a mobility friction and a potential source of internal bargaining leverage in high-skilled labor markets.

Several papers study general and firm-specific human capital accumulation. [Lazear \(2009\)](#) argues that firm-specific human capital can emerge even when skills are general, as firms require

different combinations of tasks. [Lentz and Roys \(2024\)](#) embed general and firm-specific human capital into a search model to study the relative efficiency of investing in general or firm-specific skills. [Gao et al. \(2021\)](#) and [Ma et al. \(2023\)](#) study high-skilled labor markets in which workers accumulate (general and firm-specific) human capital via learning-by-doing, and examine how this shapes mobility choices over careers. [Waldman and Zabojnik \(2025\)](#) develop a CEO labor market model in which asymmetric employer learning and firm-specific human capital are central: incumbent firms observe executive ability and firm-specific human capital, outside firms incur interview costs to learn about external candidates, and observed CEO offers signal candidate type. We build on this line of work as our model allows us to precisely identify the impact of firm-specific human capital on bargaining incentives and, in turn, on labor market outcomes.

Finally, job switching, renegotiation, bargaining, and the accumulation of general and firm-specific skills jointly generate endogenous relations between compensation, tenure, and experience that reduced-form wage models ([Mincer, 1974](#); [Abowd et al., 1999](#)) may not capture, since unobservables such as labor market competition can confound inference. Our structural approach allows us to decompose managerial compensation into components attributable to firm-specific human capital, general human capital, bargaining power, and competition, and to quantify the magnitude of the endogeneity that arises when these forces are not separately observed.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 presents the model, and Section 4 discusses estimation and identification. Section 5 reports parameter estimates and our main results concerning the drivers of managerial compensation. Section 6 examines managerial bargaining power and labor market competition, and Section 7 studies the role of firm-specific human capital in mobility and compensation. Section 8 concludes. Model proofs are in Appendix A and further estimation details in Appendix IA1.

2. Data

We estimate our model on a panel dataset covering executive careers at US public firms from 1992 to 2023. We supplement data from Execucomp with manually-collected data on when managers

first joined firms and entered the labor market, extending work from [Gayle et al. \(2015\)](#). We classify each manager in our sample as being employed in an executive (non-CEO) position, a CEO position, or unattached. Unattachment refers to managers who stop working at a firm in our sample for at least one year, but begin work at another public firm later. Managers that permanently attrit from the sample are classified as retired.⁶

Our measure of compensation is TDC1, which includes each year’s salary, bonus, stock grants, option grants, long-term incentive payouts, and other non-equity incentive payouts. Following [Taylor \(2013\)](#), we assume the annual pay level induces the manager to deploy their human capital at the firm and produce, and negotiated future incentives are incorporated into the current piece-rate. Indeed, piece-rates function as near-pure incentive contracts, directly linking compensation to output produced by the manager. We identify each labor market event for managers in our sample. Executives can be internally promoted, externally promoted, or move horizontally to be an executive at another firm. CEOs can move horizontally to other CEO positions.

Table 1 displays summary (Panel A) and labor market transitions statistics (Panel B), which are displayed conditional on employment state (executive, CEO, or unattached). CEOs receive about \$3 million at the median, \$958 thousand for executives. Wage growth is on average about 9.8% per year for CEOs, 13.7% for executives. External CEO hires are uncommon relative to internal promotions: there are 667 external CEO hires (293 poached CEOs and 374 poached executives), relative to 4,170 internal promotions. There are more executive-to-executive transitions (1,535), which prove useful as our strategy for separately identifying general and firm-specific human capital accumulation relies on pay changes around job-switching events.

As we show in Section 4, these areas of the data discipline distinct sets of structural parameters. For example, the joint distribution of wages (both levels and growth rates), experience, and tenure identifies the accumulation of general and firm-specific human capital, while observed frequencies of labor market events identify the associated structural arrival rates.

⁶We cannot perfectly separate unattachment and retirement. Indeed, any manager who initially enters unattachment but ultimately never shows up again in Execucomp is labeled as retired, so it is likely that our separation rate (retirement rate) is biased downward (upward). This has implications for the stock of managers that firms consider for unfilled positions, but does not impact wage bargaining or mobility in the model.

3. Model

3.1. Environment

The labor market consists of firms and managers. Time is continuous. For a given manager, we let t denote their total experience in the managerial labor market and $\tau \leq t$ denote experience at their current firm (i.e., tenure). At any point in time, managers may find themselves in one of three employment states: if currently employed, managers may hold a CEO position (C) or a non-CEO position (E), where we refer to the latter type of manager as an “executive.” Otherwise, managers may be “unattached” (U), in which case they do not currently hold a position in the managerial labor market.

3.1.1. Production and Human Capital

A manager’s total human capital (in logs) is given by:

$$h = a + g(t) + k(\tau) \tag{1}$$

The parameter $a \sim N(0, \sigma_a^2)$ is a (publicly-known) manager-specific ability parameter reflecting permanent differences in individual productivity. The functions $g(t)$ and $k(\tau)$ denote general and firm-specific human capital accumulated through work experience. Importantly, any time a manager leaves their firm, general human capital $g(t)$ is retained while their firm-specific capital $k(\tau)$ is not: $k(\tau)$ resets to $k(0) = 0$ upon switching firms.

At each t , each manager-firm match generates output Y , with natural logarithm y :

$$y = h + f = a + g(t) + k(\tau) + f, \quad f \sim \Psi \tag{2}$$

where $f \in [f_{\min}, \infty)$ is a firm effect summarizing firm-specific characteristics which augment managerial productivity. This may include firm size, organizational culture, proprietary technology, etc. We henceforth refer to f as *firm productivity* which, as it is firm-specific, is redrawn

from distribution Ψ any time a manager switches firms. It is convenient to define $p = f + k(\tau)$ as the manager’s *match productivity*, representing the portion of total productivity y tied to the current match, which is therefore lost upon switching firms. As we will later show, the dynamics of match productivity p play a critical role in shaping managers’ incentive to either remain in their current position or entertain outside offers.

While unattached, a manager’s tenure τ is fixed at zero and firm-specific human capital does not accumulate. It is important to note that unattachment is not synonymous with unemployment. Indeed, as discussed by [Cziraki and Jenter \(2022\)](#), unattached managers are largely those who temporarily exit employment within large, publicly-held firms in favor of other roles in, for example, government, non-profits, or privately-held firms. We therefore allow experience t , and thus general skills $g(t)$, to accumulate while unattached.

3.1.2. Labor Market Transitions and On-the-Job Search

Employed managers engage in on-the-job search and may receive alternative employment offers at any time. For executives, three possible offers may be received: internal promotion to CEO, external promotion to CEO, or a lateral move to an executive role with another firm. Internal promotion opportunities arrive at Poisson rate λ_0 , while external offers for CEO and executive roles arrive at rates λ_1 and λ_2 , respectively.⁷ Crucially, internal promotions are distinct from external transitions in that firm-specific human capital $k(\tau)$ is retained following a within-firm move, but reset to zero following a cross-firm move.

For CEOs, the only types of viable outside offers are for CEO positions at different firms; we rule out demotions *ex ante*.⁸ As in the case of executives, we assume that offers for outside CEO

⁷We take the Poisson arrival rates as a “reduced-form” of firms’ recruiting problem rather than modeling their offer-sending decisions explicitly. Unmodeled institutional constraints such as non-compete enforceability ([Shi, 2023](#)) or succession planning practices ([Cvijanović et al., 2023](#); [Geelen and Hajda, 2024](#)) will potentially load onto these arrival rates in estimation.

⁸This assumption is motivated by the data. When examining potential demotions (when a manager is the CEO at a firm in year t and a non-CEO executive in year $t + 1$) in our data, we find that a large majority are advisory positions. For example, the previous CEO stays on at the firm explicitly as an advisor to the current CEO, or implicitly by taking chairmanship of the board. However, a small number of true demotions do exist in our data. Our model is not intended to study the decision to promote a (possibly interim) CEO and subsequently demote them, so we remove these managerial spells from our estimation sample.

positions arrive at rate λ_1 . We will show that despite this assumption, the rates of executive-to-CEO and CEO-to-CEO transitions will differ in equilibrium. As in all previous cases, CEOs forgo their firm-specific capital $k(\tau)$ upon taking a position with another firm.

In addition to receiving job offers, managers may receive employment shocks inducing separation or exit from the labor market. We assume that managers retire and exit permanently at rate μ . Alternatively, at rate η manager-firm matches are dissolved and the manager enters unattached. While unattached, managers may potentially re-enter the labor market by receiving an offer for an executive or CEO position, which arrive at rates γ_E and γ_C , respectively.

3.2. Bargaining and Wage Contracts

Compensation takes the form of a piece-rate contract: managers receive an endogenous share $R \in (0, 1]$ of their marginal product Y .⁹ Consequently, log compensation is given by:

$$w = r + a + g(t) + k(\tau) + f \quad (3)$$

Piece rates $r = \log(R) \leq 0$ are determined via strategic bargaining between firms and managers, with $r = 0$ capturing the extreme case in which the manager fully extracts the surplus generated by the match.¹⁰ Throughout their careers, managers can increase their captured surplus r by leveraging competing job offers. Any time an offer arrives, a bargaining game is initiated between the manager and firm(s) bidding for their services, the outcome of which determines both who wins the manager and at what contract. The bargaining protocol is as an extension of [Cahuc et al. \(2006\)](#), the exact form of which depends on the type of offer in question: internal promotion, external promotion, or lateral move. We relegate microfoundations of the equilibrium bargaining rules to [Appendix IA2.1](#) and focus instead on the economic intuition in the next

⁹This specification follows an influential strand of the labor search literature ([Postel-Vinay and Robin, 2002](#); [Bagger et al., 2014](#)). To retain tractability, we do not explicitly model managers' dynamic moral hazard problem. Rather, the piece rate contract (3) can be taken as a reduced-form of the underlying incentive-alignment problem, where r proxies for performance-based pay.

¹⁰Equivalently, perfect competition in the market for managers implies that $r = 0$ always holds. Deviations from this perfectly-competitive benchmark stem from labor market frictions.

section. Throughout, we let $V_i(r, t, p)$ denote the discounted value of employment for a manager in a type $i \in \{E, C\}$ role given state (r, t, p) .

3.2.1. Internal Promotions

Consider a manager employed in an executive position with state (r, t, p) . Suppose their firm approaches them with a potential promotion to CEO. The firm offers piece rate r' characterizing the manager's contract were they to accept the promotion, where r' satisfies the following condition:

$$V_C(r', t, p) = V_E(r, t, p) + \beta(V_C(0, t, p) - V_E(r, t, p)) \quad (4)$$

The value to the manager of accepting promotion, $V_C(r', t, p)$, is the value of remaining in their current position $V_E(r, t, p)$ plus a share β of the additional surplus created by the new match. The parameter β , to be estimated, measures managers' bargaining power. The sharing rule (4) implies that it is in the manager's interest to accept promotion to CEO if and only if $V_C(0, t, p) > V_E(r, t, p)$. The following proposition implies a useful simplification of the internal bargaining regime:

Proposition 1. *Define $\bar{\theta}(r, p)$ as the critical level of match productivity satisfying the indifference condition:*¹¹

$$V_C(0, t, \bar{\theta}(r, p)) = V_E(r, t, p) \quad (5)$$

When $r < 0$, we have that $\bar{\theta}(r, p) < \bar{\theta}(0, p) \leq p$, which implies that:

$$V_C(0, t, p) > V_C(0, t, \bar{\theta}(r, p)) = V_E(r, t, p) \quad (6)$$

Proof. See Appendix A. ■

Proposition 1 implies that all internal promotion offers are accepted, with starting contract r' satisfying Equation (4). Internal promotions constitute a change in job title and, by virtue of managers' bargaining power, an increase in managers' capture surplus; importantly, match pro-

¹¹We suppress dependence on experience t for notational convenience.

ductivity p is unchanged. As managers assign higher value to CEO employment than executive employment (Equation (6)), internal promotions are never rejected on the equilibrium path.

3.2.2. Cross-Firm Transitions

For internal promotions, only two parties (the manager and incumbent firm) take part in negotiations. The situation is more nuanced for cross-firm job offers as negotiations entail three parties: the manager, the incumbent firm, and the poaching firm. Here, the two firms competitively bid for the manager's contract, while the manager exercises bargaining power over both. We let $f' \sim \Psi$ denote the productivity of the poaching firm, which is public information. Given an offer from a type f' firm, cross-firm bargaining has three possible outcomes which we summarize in the following proposition:

Theorem 1. *Suppose a type- i manager with state (r, t, p) receives an outside offer for a position of type j . Let f' denote the productivity of the poaching firm. The poacher wins the manager if and only if $f' > x_{ij}$, where:*

$$x_{EC} = \bar{\theta}(0, p) \tag{7}$$

$$x_{CC} = x_{EE} = p = f + k(\tau) \tag{8}$$

The incumbent retains the manager at a renegotiated contract if and only if $f' \in [\underline{\theta}_{ij}, x_{ij}]$ where the lower renegotiation bound $\underline{\theta}_{ij}(r, p)$ is defined as the solution to:

$$V_i(r, t, p) = V_j(0, t, \underline{\theta}_{ij}(r, p)) + \beta(V_i(0, t, p) - V_j(0, t, \underline{\theta}_{ij}(r, p))) \tag{9}$$

Lastly, if $f' < \underline{\theta}_{ij}$, the offer is ignored.

Proof. See Appendix A. ■

The new firm's productivity f' determines whether the manager accepts an outside offer, is retained under a renegotiated contract, or ignores the offer entirely. The external acceptance

thresholds (x_{ij}) reflect the fact that in all cases of cross-firm transitions, the match productivity of the new position must be high enough to compensate for the firm-specific capital $k(\tau)$ lost during the job transition. Additionally, the likelihood of accepting an offer declines as $k(\tau)$ increases as firm-specific human capital accumulation increases match productivity p , gradually driving up the opportunity cost of switching firms. This induces *job lock*: imperfect transferability of human capital reduces the realized rate of cross-firm transitions.

Given the acceptance criteria in Theorem 1, we next outline the bargaining outcomes in each instance. Throughout, we consider a manager employed in a position of type $i \in \{E, C\}$ with state (r, t, p) who receives an offer for a position of type $j \in \{E, C\}$. Suppose first that poacher productivity satisfies $f' > x_{ij}$. The manager accepts the offer in this case, and their new contract r' satisfies:

$$V_j(r', t, f') = V_i(0, t, p) + \beta(V_j(0, t, f') - V_i(0, t, p)) \quad (10)$$

Upon accepting the type j offer, the manager's new value $V_j(r', t, f')$ is the total surplus of their old match $V_i(0, t, p)$ plus a fraction β of the additional surplus generated by the new match. The value of accepting the competing offer reflects the fact that the manager will begin their new job spell with $k = 0$, so the match productivity at their new firm starts at f' . Further, the manager's starting rate r' depends explicitly on the productivity of their prior employer (Fee and Hadlock, 2003).

Additionally, note a subtle but important difference between the external and internal sharing rules (Equations (10) and (4)). For internal promotions, the manager's outside option is $V_E(r, t, p)$, or the value of remaining in their executive role at contract r . For external transitions, the manager's outside option is $V_i(0, t, p)$, or the value of remaining in their current role at piece rate $r = 0$ (i.e., extracting full surplus). We show in Appendix IA2.1 that competition between the two firms drives the losing firm (the incumbent in this case) to bid full surplus $V_i(0, t, p)$ which, while never accepted in equilibrium, is a viable threat point which the manager can use as leverage. Managers are thus in more favorable bargaining positions when fielding external offers (Cahuc et al., 2006).

Next, suppose the manager receives an outside offer from a firm with productivity satisfying $f' \in [\underline{\theta}_{ij}, x_{ij}]$. By Theorem 1, the manager's equilibrium response is to reject the offer and remain

with the incumbent at a renegotiated contract. This renegotiated piece rate r' satisfies:

$$V_i(r', t, p) = V_j(0, t, f') + \beta(V_i(0, t, p) - V_j(0, t, f')) \quad (11)$$

Similar to the previous case, competition between the two firms drives the losing firm's bid (the poacher in this case) up to full surplus $V_j(0, t, f')$. Again, while this is not accepted by the manager in equilibrium, the threat of acceptance induces the incumbent to favorably renegotiate their contract to incentivize retention. Thus, outside competition may still benefit the manager even if they ultimately elect not to accept the new offer. It is, however, possible that a competing offer is not threatening enough to trigger renegotiation by the incumbent. In particular, outside offers satisfying $f' < \underline{\theta}_{ij}(r, p)$ are rationally ignored, where $\underline{\theta}_{ij}(r, p)$ is an endogenous renegotiation threshold defined by Equation (9). This condition states that, given equilibrium bargaining strategies, receiving an offer from a firm with productivity $\underline{\theta}_{ij}(r, p)$ would leave the manager no better off than their current contract. As such, managers only consider offers with $f' \geq \underline{\theta}_{ij}(r, p)$.

3.2.3. Unattached Managers

Lastly, recall that unattached managers can re-enter the labor market with an executive or CEO job offer. For tractability, we follow [Bagger et al. \(2014\)](#) and assume that the value of unattachment $V_U(t)$ is equivalent to employment in the least-productive executive position: $V_U(t) = V_E(0, t, f_{min})$, implying that unattached managers accept all offers. Upon receiving an offer of type j from a type- f' firm, managers exit unattachment and begin their new role with piece rate r' satisfying:

$$V_j(r', t, f') = \beta V_j(0, t, f') + (1 - \beta) V_E(0, t, f_{min}) \quad (12)$$

3.3. Equilibrium Contracts

We next summarize the equilibrium piece rate contracts implied by the bargaining rules outlined above. We focus on piece rates set following transitions into CEO employment; discussion of the

piece rates following lateral executive moves appears in Appendix A.¹² CEOs' initial contracts will depend on the route that led them to the CEO position, whether through internal promotion, external promotion, or lateral poaching. To aid exposition, we adopt the following terminology: *horizontal* moves refer to cross-firm CEO to CEO transitions, *diagonal* moves refer to cross-firm executive to CEO transitions, and *vertical* moves refer to internal promotions. In general, CEOs' initial contracts are of the form $r_C^m(p, z)$ for $m \in \{hor, diag, vert\}$, where p is match productivity at time of negotiation and z is a jump process tracking the manager's outside option.¹³

Proposition 2. *Define the survivor function $S(x) = 1 - \Psi(x)$. Additionally, define:*

$$q(p) = 1 + \lambda_1(1 - \beta) \frac{\partial V_C}{\partial p}(0, t, p) S(p) \quad (13)$$

Let f' denote the productivity of the poaching firm and p denote the match productivity characterizing the manager's previous position. Piece rates following job switches have the following form:

$$r_C^{hor}(f', p) = -(1 - \beta) \int_p^{f'} q(x) dx \quad (14)$$

$$r_C^{diag}(f', p) = -(1 - \beta) \int_{\bar{\theta}(0, p)}^{f'} q(x) dx \quad (15)$$

$$r_C^{vert}(p) = -(1 - \beta) \int_{\bar{\theta}(r, p)}^p q(x) dx \quad (16)$$

Furthermore, for any $f' > p$, we have that:

$$r_C^{hor}(f', p) > r_C^{diag}(f', p) > r_C^{vert}(p) \quad (17)$$

¹²In Appendix A.1, we first fully derive the executive and CEO value functions, and then use these to derive equilibrium contracts.

¹³In general, z equals the match productivity of the best position the manager *could have* accepted but did not. It jumps upon receiving a meaningful offer and remains fixed in between offers. For example, consider a CEO with current match productivity $p = f + k(\tau)$. Suppose they take a CEO position at a new firm. Their outside option z has initial value p . Suppose at some point in the future, an outside CEO offer arrives from a type- f' firm, which is ultimately rejected. Following this, the CEO's outside option jumps to $z = f'$. Thus, outside options jump upon receiving a meaningful offer and remain constant otherwise. The dynamics of the renegotiation boundary $\bar{\theta}_{ij}$ ensure that z is weakly increasing. We suppress the piece rate's dependence on t for notational convenience.

Proof. See Appendix A. ■

For each type of transition, the prevailing piece rate resembles that of [Bagger et al. \(2014\)](#) with some subtle differences. First, as match productivity p increases with $k(\tau)$, starting piece rates following cross-firm transitions will increase with respect to managers' stock of firm-specific skills accumulated in their previous role. Thus, while CEOs' firm-specific capital at their previous firm has no effect on output in their new position, it directly impacts the structure of their initial contract. Namely, to offset the cost of abandoning their previously-accumulated $k(\tau)$, poaching firms must compensate managers for this loss via surplus concessions.

Further, the piece rate ordering (17) embeds several important distinctions between the path-specific bargaining protocols. First, as CEO employment is more valuable than executive employment (Proposition 1), poached CEOs have higher outside options than externally-promoted executives. As a consequence, horizontal CEO hires command higher initial piece rates than their diagonally-hired counterparts. Second, the effect (or lack) of bidding competition has differential effects on the path-specific piece rates. As discussed above, managers engaged in cross-firm negotiations hold more effective bargaining power than those approached for internal promotions: internal hires (all else equal) see lower starting pay than their externally-hired counterparts. In addition, while external hires require compensation for firm-specific skill lost during the transition, insiders do not, further widening the starting pay gap relative to external candidates.

Next, recall that upon receiving an outside offer such that $f' \in [\underline{\theta}_{ij}, x_{ij}]$, the incumbent firm retains the manager under a renegotiated piece rate. Focusing again on the case of CEOs, we summarize these renegotiated piece rates in the following proposition.

Proposition 3. *Suppose a CEO receives an outside offer from a type f' firm, where $f' \in [\underline{\theta}_{ij}, p]$. In this case, the incumbent retains the manager at the renegotiated piece rate:*

$$r_C^{reneg}(p, f') = -(1 - \beta) \int_{f'}^p q(x) dx \quad (18)$$

Proof. See Appendix A. ■

Though the renegotiated rates (18) and “switching” rates (Equations (14) and (15)) are similar in structure, the role of firm-specific skill within these two regimes is distinct. As discussed above, in the case of job switching, a portion of the poacher’s surplus concession serves the purpose of compensating the manager for their foregone firm-specific skill (Lentz and Roys, 2024). Thus, firm-specific capital enhances realized switching rents. The opposite is true in cases of renegotiation.¹⁴ Firm-specific skill accumulation gradually drives up the manager’s value of retaining employment with the incumbent. As a consequence, firm-specific skill dampens the need for incumbent firms to incentivize retention via surplus concessions; from firms’ perspective, firm-specific capital incentivizes retention “for free.”

While firm-specific human capital alters the magnitude of contractual revisions following renegotiation, it also affects the set of outside offers threatening enough to trigger renegotiation in the first place. The following proposition highlights the role of firm-specific capital on this second, extensive, margin of renegotiation:

Proposition 4. *For every transition type $ij \in \{CC, EC, EE\}$, the following holds:*

$$\frac{\partial x_{ij}}{\partial \tau} > \frac{\partial \underline{\theta}_{ij}}{\partial \tau} > 0 \quad (19)$$

Consequently, the renegotiation region $[\underline{\theta}_{ij}, x_{ij}]$ expands with tenure.

Proof. See Appendix A. ■

Proposition 4 shows that as tenure (and therefore $k(\tau)$) increases, both the acceptance x_{ij} and renegotiation thresholds $\underline{\theta}_{ij}$ shift to the right, with x_{ij} moving faster. The interval $[\underline{\theta}_{ij}, x_{ij}]$ therefore widens with tenure: conditional on receiving an offer, the likelihood of renegotiation

¹⁴To see this explicitly, consider a CEO who receives an outside offer such that $f' \in [\underline{\theta}_{CC}, f]$. Let $r_C^{reneg}(f, f')$ denote the consequent piece rate in a world with no firm-specific capital. Comparing $r_C^{reneg}(f, f')$ with (18) yields:

$$r_C^{reneg}(p, f') - r_C^{reneg}(f, f') = -(1 - \beta) \int_f^{f+k} q(x) dx < 0$$

Hence, firm-specific capital decreases the magnitude of contractual revisions following failed poaching attempts.

gradually increases. Taken together, Propositions 3 and 4 show that firm-specific human capital has opposing effects on managers' renegotiation rents. On the intensive margin, firm-specific skill accumulation decreases managers' willingness to switch firms, compressing the incremental surplus concession required in any given renegotiation. On the extensive margin, firm-specific capital expands the set of outside offers triggering renegotiation rather than switching. So, relative to a world with fully-portable skill, managers receive more frequent, but smaller, contractual revisions over a given job spell when human capital is partially firm-specific. Whether firm-specific capital is ultimately "good" or "bad" for managers' realized renegotiation rents is therefore ambiguous *ex ante*, and must be resolved empirically.

3.4. Discussion

For a manager of type $i \in \{E, C\}$, the equilibrium wage process is given by:

$$w_i = a + g(t) + k(\tau) + f + r_i^*(p, z) \quad (20)$$

where r_i^* is the realized piece rate following the manager's most recent pay negotiation. In the spirit of Mincer-AKM, equilibrium compensation admits an additive decomposition into a manager fixed effect a , an experience trend $g(t)$, a tenure trend $k(\tau)$, a firm fixed effect f , and a persistent random variable r_i^* . Despite its simple appearance, (20) embeds rich interactions between human capital accumulation, search, and bargaining. The piece rate r_i^* captures the endogenous, history-dependent impact of bargaining, outside options, and cross-firm competition on realized pay. Through the accumulation (or destruction) of firm-specific capital, selection into more productive firms, or the arrival of competing offers, r_i^* systematically varies with managerial tenure and experience. Broadly speaking, this implies that pay-tenure/experience profiles observed in the data conflate both human capital accumulation and endogenous bargaining outcomes. This underscores the value of our structural approach. By explicitly modeling the evolution of r_i^* , we cleanly recover the contribution of managerial human capital *net of* the endogenous effects of bargaining, competition, and outside options.

4. Estimation

We estimate the model via indirect inference (McFadden, 1989; Smith, 2016). We outline in this section our estimation algorithm, identification strategy, and set of target moments used to recover our structural parameters. We then discuss the estimated model’s fit. The estimation procedure goes as follows. We first compute a set of empirical moments using the estimation sample described in Section 2. Then, we simulate the model and compute an identical set of moments using the resulting simulated data set. Parameter estimates are selected to minimize the optimally-weighted distance between the empirical and simulated moments, where weights are obtained using the inverse of the variance-covariance matrix of the empirical moments, clustered at the manager level. Further details are included in Appendix IA1.

4.1. Empirical Implementation

To simulate the model, we discretize time and start with an initial cross-section of I managers at time 0, drawing each an ability $a \sim N(0, \sigma_a^2)$ and subsequently assigning them to executive and CEO positions based on the rates γ_E and γ_C . Tenure τ and experience t are initialized at zero. For each employed manager, we draw a firm heterogeneity parameter f and set the lower renegotiation bound at f_{min} .¹⁵ Following Bagger et al. (2014), we assume that firm heterogeneity follows a Weibull distribution:

$$\Psi(f) = 1 - \exp\left(-\left(\frac{f - f_{min}}{s_1}\right)^{s_2}\right) \quad (21)$$

where scale s_1 , shape s_2 , and location f_{min} are parameters to be estimated.

Labor market shocks (separation, job offers, retirement) arrive at the relevant rates at any given time. In the event of a job offer, a corresponding productivity f' is drawn and employment outcomes are realized following the discussion in Section 3.3. Managers are immediately replaced in case of retirement (which we force after 45 years of experience if it has not already occurred).

¹⁵This follows from Equation (12). As all managers are assumed unattached prior to their first position, their initial renegotiation region is $[f_{min}, f]$.

We assume that the general and firm-specific components of human capital follow cubic trends so that $g(t) = \sum_{q=1}^3 \delta_g^q t^q$ and $k(\tau) = \sum_{q=1}^3 \delta_k^q \tau^q$, where we estimate both $\{\delta_g^q\}$ and $\{\delta_k^q\}$.

Given managers' states, we compute their compensation according to (20). We augment the theoretical wage equation with additive, persistent individual shocks. For manager i , $\epsilon_{it} = \nu_e \epsilon_{it-1} + e_{it}$ captures unmodeled sources of idiosyncratic pay fluctuations.¹⁶ We assume that $e_{it} \sim N(0, \sigma_e^2)$ and jointly estimate ν_e and σ_e alongside the previously-defined structural parameters.¹⁷ We simulate the model for a total of 100 years, using the first 25 years as a "burn-in" period to allow the economy to approach a steady-state, which forms our simulated sample.¹⁸

4.2. Identification Strategy

We show there is a tight relation between the reduced-form moments of the auxiliary models and structural parameters. Our three sets of moments target different sets of structural parameters. Crucial to the success of the identification argument is separating the impacts of search frictions, imperfect labor market competition, and human capital accumulation in determining compensation and mobility over the manager's career.

4.2.1. Labor Market Mobility

Managers may transition across three labor market states: unattachment, executive employment, and CEO employment. CEO spells may end with unattachment, retirement, or a horizontal move; executive spells with an internal promotion, external promotion, or horizontal move. Unattached managers may re-enter the labor market through offers of executive or CEO employment. We ex-

¹⁶The inclusion of the process ϵ_{it} helps avoid misattributing unmodeled, match-specific components of pay to the human capital processes $g(t)$ and $k(\tau)$. For example, ϵ_{it} may capture slow-moving managerial rents stemming from entrenchment, persistent shocks to firm productivity, or noise in the grant-valuation and reporting processes.

¹⁷To avoid issues of econometric singularity, it is common practice in the empirical implementation of labor search models to add idiosyncratic noise to theoretical wage equations (see for example: [Eckstein and Wolpin, 1990](#); [Flinn, 2006](#)). These shocks are typically viewed as the natural consequence of measurement error. Strictly speaking, including the AR(1) process ϵ_{it} is not necessary in our setting, as the theoretical wage equation (20) already features unobserved heterogeneity across managers, across firms, and within employment spells. Nevertheless, we include this process in the simulated wage equation for empirical realism.

¹⁸We could in principle derive the relevant steady-state distributions analytically and initialize individual states by taking draws from these distributions. However, obtaining an analytic characterization of the steady state is intractable in our case. We therefore follow previous literature (e.g., [Bobba et al., 2021](#)) and obtain model stationarity via forward simulation.

ternally calibrate μ , the retirement rate, to 0.0619, which matches the average lifetime experience as a manager in the estimation sample of 16.16 years. To identify the arrival rates corresponding to other labor market transitions, we fit Kaplan-Meier estimates of the survivor function for each type of employment and transition event.

For example, consider horizontal CEO transitions. Let $N_C(\tau)$ be number of managers at tenure τ who can experience a horizontal CEO transition and let $M_C(\tau)$ be the number of managers that end up in a horizontal CEO transition. The Kaplan-Meier estimate of the survival function is

$$\hat{S}_C(\tau) = \prod_{s=0}^{\tau} \frac{N_C(s) - M_C(s)}{N_C(s)}.$$

The survival function estimates are defined analogously for all other labor market events. Manager survival functions for job-to-job transitions are defined over tenure, while transitions into retirement and unattachment are defined over experience. Transitions from unattachment into employment are defined over years in unattachment. For transitions between employment types and into unattachment, we match $1 - \hat{S}(5)$ and $\hat{S}(5) - \hat{S}(10)$, which give the estimated probability that a manager experiences a transition within the first five years of tenure and between the fifth and 10th year of tenure, respectively. For transitions from unattachment, we target $1 - \hat{S}(2)$, as the majority of unattached managers find managerial employment (executive or CEO) within two years in our data. Note that, as job transition probabilities in our model are endogenously related to both employer effects and tenure (which determine tenure-varying match quality), these moments also carry information about the underlying firm heterogeneity distribution and human capital accumulation parameters.

Lastly, to control the stock of CEOs in our economy, we also target the average share of managers employed in CEO positions. This ensures that the flows into CEO employment (via unattachment and internal/external promotion) and executive employment (via unattachment) are consistent with the observed structure of the managerial labor market.¹⁹

¹⁹We construct labor market transition moments from managers who experience at least one non-retirement transition throughout their career. Treating these retirement-only careers as having zero promotion or separation hazards would mechanically bias those hazards downward, as the Execucomp sample has many managers (partic-

4.2.2. Mincer Wage Regression

Our model admits a process for managerial compensation as a function of experience, tenure, manager and firm heterogeneity, and labor market competition. As such, a natural starting point for identification is an empirical model of compensation via a Mincer-AKM wage regression (Mincer, 1974; Abowd et al., 1999, i.e., AKM). We estimate, for manager i and year y :

$$w_{iy} = g_{rf}(t_{iy}) + k_{rf}(\tau_{j(i,y)}) + \zeta_{j(i,y)} + \xi_i + u_{iy}, \quad (22)$$

where w_{iy} is the manager's log total compensation and $j(i, y)$ is a matching function assigning manager i to firm j in year y . $\tau_{j(i,y)}$ is tenure at her current firm and t_{iy} is her labor market experience; both enter as cubic polynomials in our empirical model and the subscript rf signifies that the reduced-form trends for g and k may be different than the structural accumulation functions. $\zeta_{j(i,y)}$ is a firm fixed effect (tracking the manager's current employer), ξ_i is a worker fixed effect, and u_{iy} is the residual. Note that (22) does not have a structural interpretation: the empirical model suffers from misspecification as we do not observe the impact of labor market competition. Our identification strategy aims to uncover the parameters that drive this misspecification.

Experience accumulates as the manager remains in the workforce, whereas tenure resets when a manager leaves their current firm. Conditional on the firm, manager, and labor market history, variation in wages across differing levels of experience and tenure identify the parameters driving the general and firm-specific human capital accumulation functions $g(t)$ and $k(\tau)$. Targeting moments of the firm fixed effect $\zeta_{j(i,y)}$ distribution helps identify the parameters of the firm heterogeneity distribution (21). We winsorize the firm fixed effects at the 1st and 99th percentiles and target the first three moments of its empirical distribution to back out the location, shape and scale parameters of the Weibull distribution. The volatility and autocorrelation of u_{iy} help pin down σ_e and ν_e , the two parameters of the empirical error process.

ularly, non-CEO executives) who show up in the sample for a few years and then attrit, which relates to compensation disclosure (i.e., reporting only the top 5 highest paid employees). This selection potentially over-represents managers with active mobility histories, so the targeted CEO employment share is higher than what we would see in the full Execucomp sample, about 34% in the estimation vs. 20% in the full sample.

4.2.3. Within- and Across-Job Wage Growth

Focusing on managers with at least two years of consecutive employment, we consider the following empirical model of managerial wage growth within- and across-jobs:

$$\Delta w_{iy} = \Delta g_{rf}(t_{iy}) + \Delta k_{rf}(\tau_{iy}) + \zeta_{j(i,y)} + \zeta_{j(i,y-1)} + \Delta u_{iy}, \quad (23)$$

where Δg_{rf} and Δk_{rf} represent the change in the tenure and experience cubic polynomials year-over-year. We include current and previous firm effects to control for firm-specific wage premia at job switch. Importantly, because tenure resets upon switching firms, Δg_{rf} and Δk_{rf} provide identification of the human capital trends as we control for contemporaneous changes in firm wage premia. We also target the volatility and autocorrelation of Δu to help identify σ_e and ν_e .

The difference between the level and first-differenced wage (22 and 23, respectively) convey information about the bargaining power parameter β , which controls the response of compensation to job changes (a change in employer type f) and to contract renegotiations (a change in the piece rate r induced by job offers). Variation in compensation across both tenure and experience profiles and around job events (switches/renegotiations) jointly inform β .

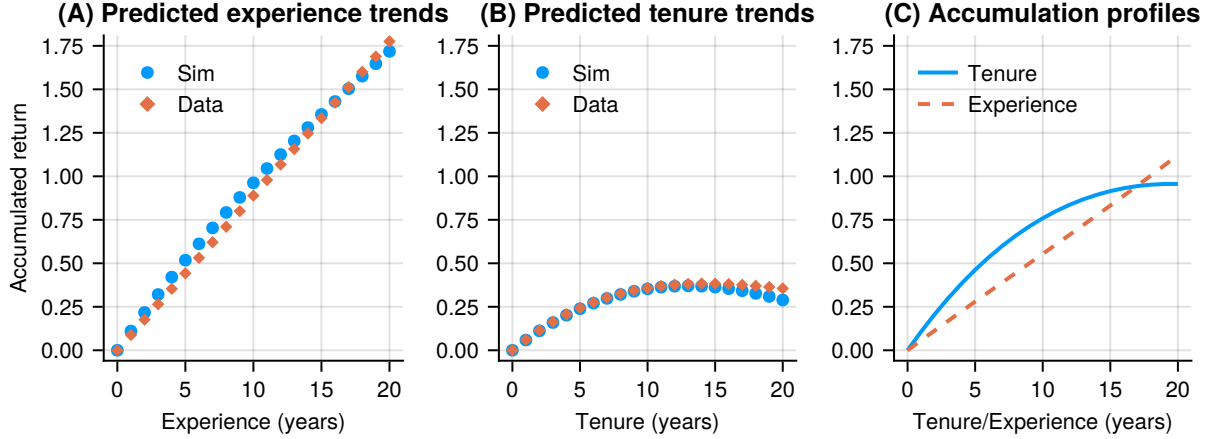
4.3. Model fit

Labor market mobility. Panel A of Table 2 reports job transition probabilities based on the Kaplan-Meier estimates and the share of CEO. In both the simulated and empirical sample, internal CEO hires are by far the most common transitions into CEO employment, followed next by external promotions and lastly by lateral CEO transitions. Our model also generates a near-perfect match of the total share of managers employed in CEO positions. Overall, our model matches the key characteristics of managerial employment dynamics.

Auxiliary regressions. Table 2 Panel B reports the moments of the Mincer wage regression as in (22) and the wage growth regression as in (23). Our model closely matches the moments of the Mincer wage regression. The simulated coefficients on the quadratic and cubic terms in

Figure 1. Cumulative wage-experience and wage-tenure profiles

This figure displays cumulative wage-experience and wage-tenure profiles. Panel A and Panel B compare the experience and tenure profiles based on the Mincer wage regression (22), by using the tenure and experience polynomial coefficients shown in Table 2, Panel B. Panel C plots the structurally-estimated tenure and experience accumulation profiles based using the tenure and experience cubic coefficients in Table 3.



the tenure and experience polynomials differ slightly from the data, but their combined effects are similar (see Figure 1). In the wage growth regression, residual variance and covariance are well-matched, and the same logic applies to the tenure and experience growth terms as in the wage growth regression.

Experience and tenure compensation profiles. Figure 1 shows the reduced-form and structural tenure and experience profiles for wages. Panel A and Panel B compare the reduced-form experience and tenure profiles based on the Mincer wage regression (22). The profiles align very closely. Panel C plots the structural tenure and experience accumulation profiles based on the model wage process (20), using the estimated parameters. As job moves reset tenure to zero while simultaneously raising wages through labor market competition, the reduced-form experience (tenure) profile is biased upwards (downwards) relative to its structural counterpart.

5. Estimation Results

5.1. Structural parameter estimates

Parameter estimates are reported in Table 3. Regarding transitions into CEO employment, our estimates of λ_0 (0.0279) and λ_1 (0.0057) imply that the arrival rate of internal CEO offers is nearly five times that of external offers. Several factors may explain this disparity, including the prevalence of non-compete agreements (Shi, 2023), external search costs (Geelen and Hajda, 2024), tournament incentives (Agrawal et al., 2006), or non-pecuniary preferences for internal candidates (He and Schroth, 2024; Capron et al., 2024). While this gap in offer rates is large, the gap in actual internal and external CEO transitions is further amplified by firm-specific skill.

Our estimates of the human capital processes $g(t)$ and $k(\tau)$ (Figure 1 Panel C) suggest that the accumulation of general human capital is near linear with respect to experience, whereas firm-specific human capital plateaus after roughly 15 years of tenure. The trajectory of *total* human capital $h(t, \tau)$ is contingent on managers' career path. As an example, consider two managers of equal ability (a), both with 10 years of experience. Total human capital will be about 36% higher for the manager who stayed in the same firm for 10 years relative to one who switched firms in year 5. Thus, job stability enhances human capital accumulation, raising the opportunity cost of switching firms and contributing in part to the empirically low rate of cross-firm mobility.²⁰

Our estimates of the distributional parameters s_1 , s_2 , and f_{min} imply that heterogeneity in firm productivity (f) is substantial, suggesting that much of the observed dispersion in managerial pay is driven by permanent firm-specific factors. Further, while we do not impose any direct dependence of f on experience, the equilibrium distribution of firm productivity varies systematically with experience. We illustrate this in Figure B.1, which plots the simulated distributions of firm productivity separately for different levels of managerial experience. In terms of stochastic

²⁰In addition, Table B.1 reports a variance decomposition of total accumulated human capital into its general and firm-specific components. In aggregate, 55% of variation in total accumulated human capital is driven by general skill, though this proportion varies significantly with both experience and tenure. The share of the variation in total human capital explained by firm-specific skills is generally higher early on in managers' careers: 65% of variation is driven by firm-specific skill among those with less than five years in the labor market, and this share drops to 41% for those between 10 and 25 years of experience.

dominance, firm productivity increases with experience, indicating that managers endogenously sort into increasingly productive firms as their careers progress. Hence, positions in the most productive firms will on average be occupied by managers with a large stock of general skills.

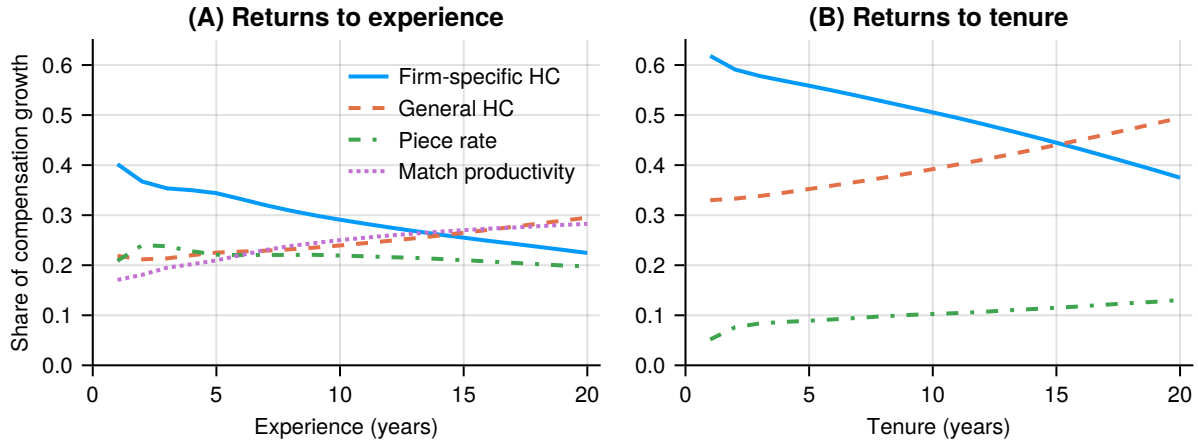
5.2. Decomposing Managerial Compensation Growth

To understand the drivers of compensation growth over careers, in Figure 2, we decompose both experience and tenure-driven growth in managerial compensation. The key drivers of growth are human capital accumulation, both general and firm-specific, growth in managers' extracted surplus (i.e., piece rates), and transitions into more productive firms. As firm productivity is fixed within position, the contribution of productivity to pay growth over tenure is zero by construction. As such, we omit firm productivity from the tenure decomposition. Starting first with the experience decomposition in Panel A, we begin by computing average CEO compensation growth for each year of experience. We then scale average growth of each experience-varying component of CEO pay by total compensation growth to compute the shares attributable to each over a manager's career. Panel B displays the same decomposition within-firm, again omitting firm productivity as an explanatory mechanism.

When explaining the returns to experience, Panel A suggests that general and firm-specific human capital together explain over half of compensation growth. Firm-specific human capital is the dominant force early in managerial careers: for managers with 5 years of experience, roughly 35% and 20% of total compensation growth is explained by firm-specific and general human capital accumulation, respectively. Later in the career cycle, the relative importance of these two channels flips: for managers with 20 years of experience, roughly 22% and 30% of compensation growth is explained by firm-specific and general human capital, respectively. The share of pay growth stemming from piece-rate improvements is stable, hovering around 22%, but moderately decreases with experience. This suggests that early-career managers have the most scope for piece rate improvements, while compensation gains later in the career stem increasingly from other channels. Conversely, the share attributable to firm productivity growth steadily increases

Figure 2. Managerial Compensation Growth Decomposition

This figure shows accumulated wage growth for CEOs over tenure and experience. We use (20) to decompose accumulated wage growth into changes in the piece rate, general and firm-specific human capital, and firm productivity (which does not play a role for tenure).



with experience. For the average mid-career managers with 15-20 years of experience, transitioning into higher productivity firms is among the top drivers of compensation growth.

When explaining the returns to tenure (Panel B of Figure 2), human capital accumulation is by far the largest driver of pay growth. As with experience, firm-specific skills contribute the most to growth early in tenure, while general skills become the front runner in late years of tenure; for managers with 5 (20) years of tenure, roughly 56% (38%) and 35% (50%) of compensation growth is respectively explained by firm-specific and general human capital accumulation. The contribution of piece rate growth is modest and remains stable throughout CEO tenure. The stability of the within-position piece rate contribution suggests that in spite of the job-lock effect, the influence of outside competition on CEO compensation (i.e., renegotiation) is persistent. However, comparing this with the piece rate contribution in the returns to experience, it is clear that within-position piece rate renegotiations are less consequential than cross-position piece rate increases due to job transitions.

5.3. Decomposing Contractual Returns

Figure 2 suggests that growth in managerial piece rates, what we call *contractual returns* for brevity, are an important source of managerial compensation growth. Over the career cycle, contractual returns are realized through two channels: (1) transitioning into a firm offering a more generous piece rate or (2) renegotiation with the incumbent firm. Our model allows a precise decomposition of contractual returns into these two components.

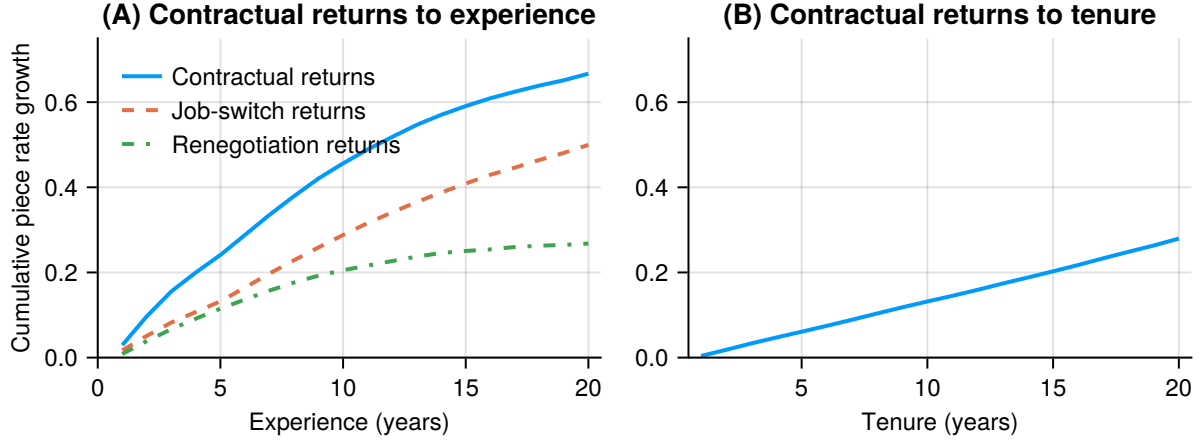
Figure 3 Panel A decomposes contractual returns over the experience profile, showing that surplus gains from switching firms account for about two thirds of contractual returns to labor market experience, while renegotiation explains the remainder. Thus, while job switches are infrequent, the returns to the occasional switch are quite potent (Fee and Hadlock, 2003), thus comprising the majority of contractual returns over the career cycle. Renegotiations following failed poaching attempts, while more frequent, yield smaller contractual gains. Panel B similarly displays contractual returns over (contiguous) tenure at the incumbent firm. An important, if perhaps obvious, point is that contractual returns over tenure come entirely from renegotiation, so contractual returns and renegotiation returns overlap perfectly, reaching about 25% by the manager's 20th year of tenure. As such, renegotiation with incumbents is the key driver of within-firm managerial surplus capture dynamics.

Upon receiving a competing offer, whether it results in job switching or renegotiation, two key channels determine the magnitude of the consequent contractual adjustment. First, the prevailing piece rate will depend explicitly on the productivity of the losing bidder. In cases of job switching, the incumbent loses and the poacher entices the manager to switch with a surplus concession which, importantly, increases in the incumbent's productivity. This is reminiscent of Fee and Hadlock (2003), who show empirically that hiring grants used to attract outside managers increase with respect to the productivity of managers' previous employer. The same intuition holds in cases of renegotiation. Here, the poacher loses and the incumbent retains the manager via a surplus concession, which again increases with respect to the poacher's productivity.

Second, contractual adjustments are influenced by the manager's stock of firm-specific human

Figure 3. Contractual Returns Decomposition

This figure displays a decomposition of cumulative piece rate growth across labor market experience and tenure. The solid (blue) line displays cumulative piece rate growth across labor market experience (tenure in Panel B). The dashed (orange) line displays contractual returns due to job-switches (i.e., accumulated piece rate growth due to switching firms). The dashed-dotted (green) line displays contractual returns due to renegotiations (i.e., accumulated piece rate growth due to renegotiations with the current firm).



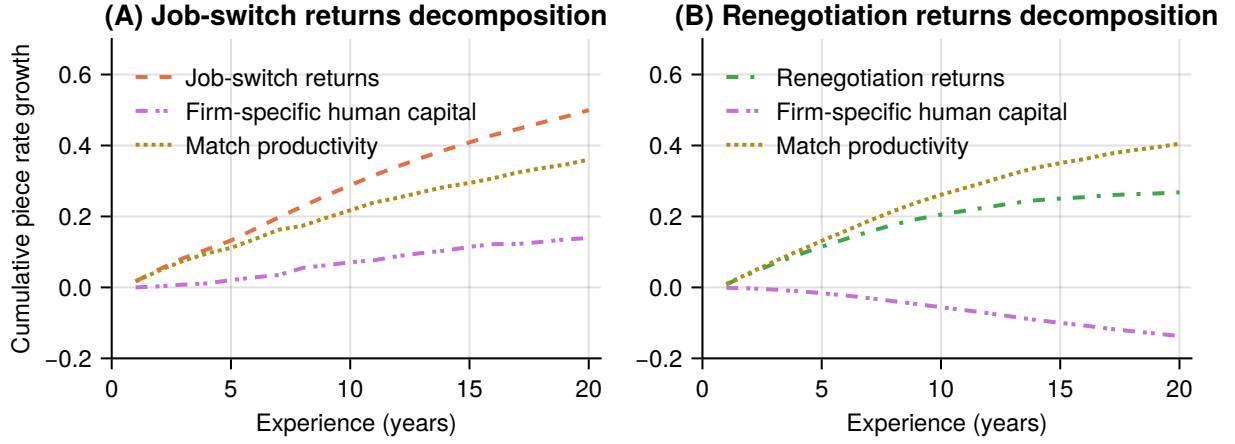
capital. As discussed in Section 3, firm-specific skill accumulation induces job lock, consequently increasing the cost of incentivizing cross-firm transitions while decreasing the cost of incentivizing retention. Thus, firm-specific skill enhances managers' captured surplus in the rare instance of switching firms, while dampening captured surplus in cases of renegotiation.²¹

The contractual returns to both job switching and renegotiation can be precisely decomposed into these two mechanisms; which we illustrate in Figure 4. Starting with the returns to job switching, Panel A splits the associated contractual returns into a firm productivity and firm-specific skill component. The yellow curve represents the cumulative growth in managerial piece rates stemming purely from incumbent productivity. Firm-specific human capital induces a notable upward shift in these switching rents; for managers with 20 years of experience, roughly 25% of switching rents are attributable to compensation for foregone firm-specific human capital. Panel B of Figure 4 similarly decomposes the contractual returns to renegotiation. In contrast with the job-switching case, firm-specific human capital has a strictly negative effect on realized

²¹This follows from Proposition 3: as firm-specific skill accumulates, the incumbent match becomes more valuable, so a given outside offer poses a weaker threat and can be countered with a smaller contractual concession, thereby reducing the surplus gained in renegotiation due to firm-specific capital.

Figure 4. Firm-Specific Human Capital and Contractual Returns

Taking the “Job-switch returns” curve in Panel A of Figure 3 as a starting point, Panel A of Figure 4 further decomposes these returns into a firm productivity and human capital component. Analogously, Panel B decomposes the “Renegotiation returns” curve in Panel A of Figure 3.



renegotiation rents; for managers with 20 years of experience, firm-specific capital decreases these accumulated renegotiation rents by roughly 12 percentage points.

5.4. Decomposing Dispersion in Managerial Compensation

To complement the within-career decompositions of managerial pay growth presented above, we decompose the cross-sectional dispersion in observed pay. Specifically, we conduct a variance decomposition of managerial compensation to assess the relative importance of human capital, firm characteristics, and bargaining outcomes. Table 4 displays the decomposition. We condition on permanent managerial ability and focus on the dynamic drivers of pay dispersion. Column 1 shows the decomposition for the entire simulated sample, whereas columns 2-4 and 5-7 provide the decomposition across different levels of experience and tenure. Column 1 shows that human capital accounts for about 16% of total wage dispersion, with firm-specific human capital contributing roughly six percentage points.²² Variation in piece rates alone explains roughly

²²While human capital is an important driver of wage growth, its contribution to dispersion is somewhat lower because human capital accumulates smoothly over experience and tenure, whereas other components such as piece rates are driven by discrete events, introducing sharp variation across managers that human capital does not generate. Table B.1 shows that general human capital accounts for about 55% of dispersion in accumulated human capital, consistent with each components contribution to wage dispersion.

one-third of the overall variance, underscoring the importance of (unobserved) labor market competition in shaping managerial compensation.

The covariance between general human capital and piece rates is positive and accounts for 10.5% of the total dispersion which, in line with the growth decompositions above, suggests that managers extract increasing amounts of surplus as experience accumulates. Further, as the competitive channels embedded in the piece rate are unobservable, this implies some degree of upward bias in reduced-form estimates of the returns to experience. The covariance between firm-specific human capital and piece rates is negative and grows in magnitude as labor market experience increases. This suggests that for managers with high levels of experience, those with the highest piece rates tend to have lower levels of firm-specific human capital. In other words, the managers extracting the most surplus tend to be those who have switched firms frequently. This echoes the findings of Figure 3, which illustrated that job switching, as opposed to renegotiation, yield the largest increases in managerial surplus.

Overall, this decomposition illustrates that the dynamics of competition (captured by the piece rate) drive a significant amount of the cross-sectional dispersion in managerial pay. Beyond the direct effect of competition on pay, the influence of competition also varies systematically with managers' observed tenure and labor market experience. This has important implications for empirical research on managerial pay. Namely, as the effect of competition is unobserved, time-varying, and manager-specific, reduced-form Mincer-style regressions necessarily yield biased estimates of the effects of experience and tenure on pay, which can be seen in Figure 1. In the next section, we further analyze the effect of competition on observed pay, focusing attention on the role of managerial bargaining power.

6. Managerial Bargaining Power and Labor Market Competition

Our estimate of β (0.2640) suggests a lower level of managerial bargaining power than what has been previously reported in related literature. In fact, our findings suggest that managers' ex ante

bargaining power is on par with workers in standard labor markets.²³ However, managers’ *effective* bargaining power is much higher than this ex ante baseline. In our model, managers capture rents through both pure bargaining power and cross-firm competition. Hence, bargaining power and competition work in tandem to increase compensation. Critically, modeling the influence of outside competition on managerial pay-setting allows us to decouple the effects of competition and bargaining power as determinants of CEOs’ captured surplus (Cahuc et al., 2006).

This is particularly important in the context of corporate executives. The high observed share of managerial rents is often taken as evidence of managers exercising undue influence on the pay-setting process (Bebchuk and Fried, 2006). Our results add nuance to this interpretation, suggesting that managers’ captured surplus is attributable in part to well-functioning labor market competition. In this section, we present a decomposition of managers’ observed share of surplus into components attributable to labor market competition and pure bargaining power. The results demonstrate that the influence of competition on managers’ captured surplus is substantial, and varies significantly with tenure, experience, and CEO type (i.e. internal/external hire).

6.1. Decomposition of Managerial Surplus

To disentangle the compensation effects of competition and bargaining power, we follow Cahuc et al. (2006) and derive a “naive” estimate of bargaining power in an alternative version of the model with no on-the-job search, and hence no cross-firm competition. By comparing this alternative estimate of bargaining power to our main estimate in Table 3, we can then compute the share of managers’ captured surplus attributable to competitive forces as opposed to pure bargaining power. For CEOs, this exercise amounts to setting on-the-job search parameter λ_1 to zero, which forces the outside option to unattachment and in turn ascribes any surplus in excess of unattachment value to pure bargaining power.²⁴ For a CEO in a type f firm with tenure τ and

²³For example, Taylor (2013) estimates that CEOs capture about 50% of surplus given upside news about their ability; Page (2018); Barry (2024) find that Boards place large weights on CEO utility, pushing CEO pay above value-maximizing levels. Conversely, Bagger et al. (2014), using French administrative data, find that *worker* bargaining power is about 0.3, similar to our estimates for managers.

²⁴Recall that in our model, unattachment has the same value as employment in the least productive firm, f_{min} .

experience t , eliminating on-the-job search implies the following log wage equation:

$$w = b(p, z)(f + k(\tau)) + (1 - b(p, z))f_{min} + a + g(t), \quad (24)$$

where $b(p, z)$ measures the CEO's observed share of surplus. In the absence of modeling on-the-job search, $\mathbb{E}[b(p, z)]$ would be our estimate of CEO bargaining power. We can relate this to bargaining power in the main model (β) by setting equal the structural wage equation and the counterfactual wage equation (24). Doing so yields:

$$b(p, z) = 1 + \frac{r(p, z)}{p - f_{min}} \quad (25)$$

Note that $b(p, z) \in [\beta, 1]$,²⁵ so the bias that arises when neglecting competition is weakly positive. We can then decompose average total surplus b into a bargaining component and competition component using the simple identity:

$$1 = \underbrace{\mathbb{E}\left[\frac{\beta}{b(p, z)}\right]}_{\text{CEO bargaining power}} + \underbrace{\mathbb{E}\left[\frac{b(p, z) - \beta}{b(p, z)}\right]}_{\text{labor market competition}} \quad (26)$$

We estimate $b(p, z)$ in our simulated sample and report results in Table 5. In the full sample, our naive estimate of managerial bargaining power (i.e., realized surplus capture) is 49.5%, in line with previous estimates in the literature (Taylor, 2013). Using the decomposition in (26), this implies that roughly 46.6% of the observed share of CEO surplus stems from labor market competition as opposed to pure bargaining power. Thus, despite the low empirical rate of CEO cross-firm mobility, the influence of competition on CEO pay is substantial.

Additionally, we report a number of subsample estimates of $b(p, z)$ to illustrate that this bias varies systematically across observables. Panel A of Table 5 reports estimates separately across

²⁵If $r(p, z) = 0$, then $b = 1$. The minimum value of $r(p, z)$ arises when the true λ_1 equals zero, where

$$r(p, f_{min}) = -(1 - \beta)(f + k(\tau) - f_{min})$$

and $b = \beta$.

CEO type, namely whether the CEO was promoted from within or poached from outside. Panels B and C report estimates across the distributions of experience and tenure, respectively. Starting with the CEO type subsamples, this bias is largest among poached CEOs; competition explains roughly 66.9% (47.2%) of surplus for external (internal) CEO hires. Attempting to poach a manager initiates a bidding war with the incumbent, and this competition can be leveraged by the manager to negotiate a more favorable contract. This is distinct from internal bargaining, where there is no direct influence of outside competition.

We also see heterogeneous effects of labor market competition on captured surplus across the distributions of experience and tenure. Panel B of Table 5 shows that the share of managerial surplus explained by labor market competition increases with experience. This follows from the history-dependence of piece-rate contracts. The impact of a one-off bidding war has a persistent effect on managers' compensation throughout their careers. As experience increases, these extracted benefits accumulate and comprise an increasing share of their total captured surplus. Likewise, competition explains a growing share of surplus as tenure increases, and the rationale is similar to the case of experience. As tenure progresses, managers receive a growing number of outside offers triggering renegotiations with their incumbent employers. This gradually increases their share of total surplus, despite their inherent bargaining power (β) remaining constant.

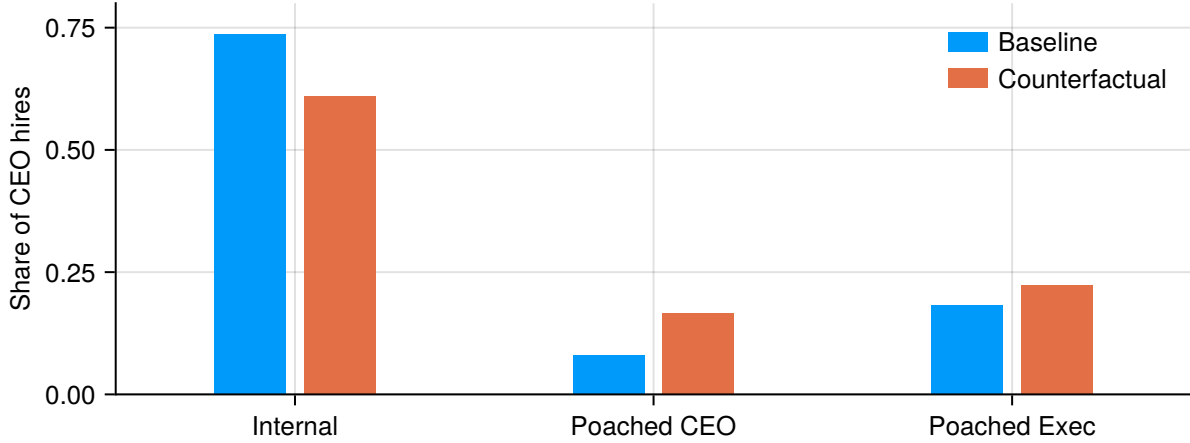
7. Firm-Specific Human Capital, Mobility, and CEO Rents

The previous section demonstrates that the influence of competition on observed CEO pay is substantial. This result may be surprising: low managerial mobility may suggest that competition plays a small role. If the majority of CEOs work for only one firm over their career, how can cross-firm competition have such a large impact on compensation? This apparent puzzle is the focus of the section, which we resolve by illustrating the impact of firm-specific human capital on both cross-firm mobility and CEOs' ability to capture rents.

We do this by comparing CEO compensation and employment outcomes in our estimated model with those in a counterfactual version that eliminates firm-specific human capital, isolat-

Figure 5. Firm-specific human capital and internal vs. external CEO hiring

This figure displays baseline and counterfactual (with $k = 0$) proportions of internal CEO promotions, externally poached CEOs and externally poached non-CEO executives.



ing the impact of firm-specific skills on both mobility and wage bargaining. We first discuss how much firm-specific human capital explains observed shares of internal and external CEO hiring. We then examine how managerial bargaining outcomes and the persistent effects of competition are affected by human capital. Although firm-specific human capital restricts managerial mobility, it does not fully insulate pay from outside competition. Instead, it shifts CEO surplus extraction from external job transitions toward internal renegotiation.

7.1. Human Capital and Transitions into CEO Employment

In Figure 5, we plot the baseline and counterfactual shares of CEOs hired from within the firm, poached from CEO positions, or poached from executive positions. Relative to the counterfactual, firm-specific human capital increases the proportion of internally-hired CEOs while decreasing the share of external hires. This results from job lock: as firm-specific human capital accumulates, the opportunity cost of switching employers gradually increases. As managers' cost of switching firms is high, we see the vast majority of CEOs being promoted from within. Note however, that even in the counterfactual, internal promotions remain the most common route to CEO employment. Thus, while firm-specific human capital accumulation explains a portion of firms' "preference" for hiring internally, other factors appear to play a material role.

Two factors explain the remaining gap in the internal and external CEO appointment rates. First, our estimates imply that $\lambda_0 > \lambda_1$, indicating that insiders are more likely to receive CEO offers in the first place. As these parameters are exogenously given in our model, whether this gap in arrival rates is the product of inefficient preference for internal candidates (He and Schroth, 2024; Capron et al., 2024) or the efficient bypassing of external search costs (Geelen and Hajda, 2024) is outside the scope of our analysis.

What our model is well-positioned to say is that this disparity in hiring rates is partly an artifact of labor market competition. When attempting to hire externally, firms must contend with the manager’s incumbent employer. This decreases the likelihood of a successful poach, as incumbent employers have the capacity to dissuade managerial exit via contract renegotiations. In the case of internal promotions, negotiations are free from such competitive pressure, representing a “path of least resistance” to the top of the job ladder. Hence, the high share of internal CEO hires is not entirely the byproduct of market frictions, but an outcome to be expected in the face of labor market competition.

7.2. Firm-Specific Human Capital and Managerial Rent Extraction

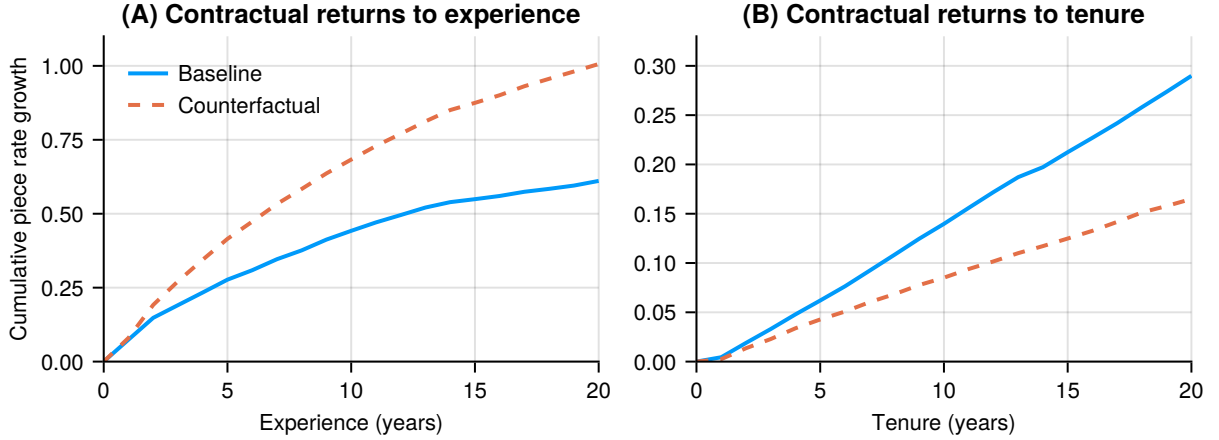
Beyond its impact on mobility, firm-specific human capital reshapes competition for top managers through two tightly linked margins. The first is an extensive renegotiation margin; Proposition 4 implies that firm-specific skill accumulation widens the renegotiation band. Put differently, conditional on receiving an outside offer, firm-specific capital increases the likelihood that the offer results in renegotiation. The second is an intensive margin effect; Proposition 3 implies that firm-specific capital attenuates CEOs’ surplus gains stemming from these renegotiations. Relative to a world with fully-portable skill, firm-specific capital thus results in more frequent, but smaller, increases in CEO surplus over their tenure in a given position (all else equal).

In Figure 6, we illustrate the aggregate effect of firm-specific skill on CEO rent extraction;²⁶ Panel A plots average piece rate growth over experience while Panel B does the same over tenure

²⁶We focus on CEOs in Figure 6 to remain consistent with the mobility analysis in Figure 5.

Figure 6. Firm-specific human capital and CEO rent extraction

Panel A plots average piece rate growth over the first 20 years of managers' careers. Specifically, for a given manager, letting $r(0)$ and $r(t)$ respectively denote their piece rate in the initial and t^{th} year of their career, we plot the average of $r(t) - r(0)$ across all managers. Panel B is analogous, where we instead define piece rate growth over tenure in a given position. In both Panels, we plot piece rate trends in both the baseline case and the counterfactual case with no firm-specific human capital.



at a firm. Focusing on the experience profile, eliminating firm-specific human capital removes job lock and facilitates cross-firm mobility, shifting extracted contractual gains from incumbents to poaching firms. By 20 years of experience, the baseline curve is approximately 40 percentage points below the counterfactual, highlighting the impact of job lock.

Over tenure, we see the opposite pattern: Panel B shows that firm-specific skill roughly doubles observed piece rate growth relative to the counterfactual. As discussed above, firm-specific skill accumulation compresses CEOs' surplus gain from any given renegotiation episode (Proposition 3), while simultaneously widening the renegotiation band (Proposition 4). Panel B illustrates that the net effect of these two opposing forces is positive: within a continuing match, incumbents more frequently concede incremental increases in the piece rate in order to retain the CEO's accumulated match value, generating faster growth in CEO rents over tenure even as cross-firm mobility declines. In other words, as firm-specific capital increases CEOs' private cost of switching firms, outside offers that would otherwise trigger a move are more often met by incumbent matching. This leads CEOs to remain in the same match, extracting rents through repeated renegotiation, as opposed to transitioning into more productive firms.

8. Conclusion

We present a structural model of the managerial labor market that quantifies the importance of general and firm-specific human capital accumulation, managerial bargaining power, and labor market competition. Our paper makes three primary empirical contributions. First, we measure the relative importance of general and firm-specific human capital in managerial skill over the manager’s career, and show that firm-specific human capital lowers cross-firm mobility (less job-switching). Second, we decompose realized managerial surplus capture into contributions from pure bargaining power and labor market competition, showing that competition effects explain roughly half of managers’ observed surplus capture. Third, we show that firm-specific human capital reshapes managerial surplus capture over the course of a career by changing the relative value of the manager’s inside and outside options.

Our work has important implications for the corporate finance literature. First, our decomposition of managerial human capital is crucial for understanding the CEO labor market, particularly the literature focusing on general CEO skill ([Gabaix and Landier, 2008](#); [Murphy and Zabojnik, 2007](#); [Frydman, 2019](#); [Cziraki and Jenter, 2022](#); [Barry et al., 2025](#)). Second, our decomposition of realized CEO surplus is important for understanding the role of CEO bargaining power and agency frictions in determining CEO compensation. We show that, in a frictional labor market in which lateral CEO transitions are rare, CEOs’ captured surplus can still be heavily influenced by competition in the labor market. In models without an explicit managerial labor market, estimates of agency frictions which induce high CEO pay may be over-stated.

References

- Abowd, J. M., Kramarz, F., and Margolis, D. N. (1999). High wage workers and high wage firms. *Econometrica*, 67(2):251–333.
- Adenbaum, J., Babalievsky, F., and Jungerman, W. (2024). Learning on the job. *Working Paper*.
- Agrawal, A., Knoeber, C. R., and Tsoulouhas, T. (2006). Are outsiders handicapped in CEO successions? *Journal of Corporate Finance*, 12(3):619–644.
- Bagger, J., Fontaine, F., Postel-Vinay, F., and Robin, J.-M. (2014). Tenure, experience, human capital, and wages: A tractable equilibrium search model of wage dynamics. *The American Economic Review*, 104(6):1551–1596.
- Barry, J. W. (2024). Shareholder voice and executive compensation. *Working Paper*.
- Barry, J. W., Graham, J. R., Kim, D., and Kim, H. (2025). Executive mobility in the United States, 1920 to 2023. *Working Paper*.
- Bebchuk, L. A. and Fried, J. M. (2003). Executive compensation as an agency problem. *Journal of economic perspectives*, 17(3):71–92.
- Bebchuk, L. A. and Fried, J. M. (2006). Pay without performance: Overview of the issues. *Academy of Management Perspectives*, 20(1):5–24.
- Bobba, M., Flabbi, L., Levy, S., and Tejada, M. (2021). Labor market search, informality, and on-the-job human capital accumulation. *Journal of Econometrics*, 223(2):433–453. Annals issue: Implementation of Structural Dynamic Models.
- Cahuc, P., Postel-Vinay, F., and Robin, J.-M. (2006). Wage bargaining with on-the-job search: Theory and evidence. *Econometrica*, 74(2):323–364.
- Capron, A., Hamilton, B., Hincapié, A., and Lyman, N. (2024). Insider CEOs: Lucky or good? *Working Paper*.
- Coles, J. L., Daniel, N. D., and Naveen, L. (2014). Co-opted boards. *The Review of Financial Studies*, 27(6):1751–1796.
- Custódio, C., Ferreira, M. A., and Matos, P. (2013). Generalists versus specialists: Lifetime work experience and chief executive officer pay. *Journal of Financial Economics*, 108(2):471–492.
- Cvijanović, D., Gantchev, N., and Li, R. (2023). CEO succession roulette. *Management Science*, 69(10):5794–5815.
- Cziraki, P. and Jenter, D. (2022). The market for CEOs. *Working Paper*.
- Duffie, D. and Singleton, K. J. (1993). Simulated moments estimation of markov models of asset prices. *Econometrica*, 61(4):929–952.

- Eckstein, Z. and Wolpin, K. I. (1990). Estimating a market equilibrium search model from panel data on individuals. *Econometrica*, 58(4):783–808.
- Edmans, A., Gabaix, X., and Jenter, D. (2017). Executive compensation: A survey of theory and evidence. *The handbook of the economics of corporate governance*, 1:383–539.
- Eisfeldt, A. L. and Papanikolaou, D. (2013). Organization capital and the cross-section of expected returns. *The Journal of Finance*, 68(4):1365–1406.
- Fee, C. E. and Hadlock, C. J. (2003). Raids, Rewards, and Reputations in the Market for Managerial Talent. *The Review of Financial Studies*, 16(4):1315–1357.
- Felli, L. and Harris, C. (1996). Learning, wage dynamics, and firm-specific human capital. *Journal of Political Economy*, 104(4):838–868.
- Flinn, C. J. (2006). Minimum wage effects on labor market outcomes under search, matching, and endogenous contact rates. *Econometrica*, 74(4):1013–1062.
- Frydman, C. (2019). Rising Through the Ranks: The Evolution of the Market for Corporate Executives, 1936–2003. *Management Science*, 65(11):4951–4979.
- Gabaix, X. and Landier, A. (2008). Why has CEO pay increased so much? *The quarterly journal of economics*, 123(1):49–100.
- Gao, J., Wang, W., and Wu, Y. (2021). Human capital portability and worker career choices: Evidence from M&A bankers. *Available at SSRN 3724657*.
- Gayle, G.-L., Golan, L., and Miller, R. A. (2015). Promotion, turnover, and compensation in the executive labor market. *Econometrica*, 83(6):2293–2369.
- Geelen, T. and Hajda, J. (2024). Succession. *Working paper*.
- He, L. and Schroth, E. J. (2024). Why are CEOs rarely succeeded by external candidates? *Available at SSRN 4728228*.
- Jovanovic, B. (1979). Firm-specific capital and turnover. *Journal of Political Economy*, 87(6):1246–1260.
- Kecht, C., Lizzeri, A., and Saidi, F. (2025). Aging at the very top. *Working paper*.
- Lazear, E. P. (2009). Firm-specific human capital: A skill-weights approach. *Journal of political economy*, 117(5):914–940.
- Lentz, R. and Roys, N. (2024). Training and search on the job. *Review of Economic Dynamics*, 53:123–146.
- Lustig, H., Syverson, C., and Van Nieuwerburgh, S. (2011). Technological change and the growing inequality in managerial compensation. *Journal of Financial Economics*, 99(3):601–627.
- Lyman, N. (2024). Reputation, entrenchment, and dynamic managerial incentives. *Working Paper*.

- Ma, S., Wang, W., and Wu, Y. (2023). Steering labor mobility through innovation. *Working Paper*.
- McFadden, D. (1989). A method of simulated moments for estimation of discrete response models without numerical integration. *Econometrica: Journal of the Econometric Society*, pages 995–1026.
- Mincer, J. (1974). Schooling, experience, and earnings. *Human Behavior & Social Institutions* No. 2.
- Moscarini, G. (2005). Job matching and the wage distribution. *Econometrica*, 73(2):481–516.
- Murphy, K. J. and Zabojnik, J. (2004). CEO pay and appointments: A market-based explanation for recent trends. *American economic review*, 94(2):192–196.
- Murphy, K. J. and Zabojnik, J. (2007). Managerial capital and the market for CEOs. *Available at SSRN 984376*.
- Page, T. B. (2018). CEO attributes, compensation, and firm value: Evidence from a structural estimation. *Journal of Financial Economics*, 128(2):378–401.
- Parrino, R. (1997). CEO turnover and outside succession a cross-sectional analysis. *Journal of Financial Economics*, 46(2):165–197.
- Postel-Vinay, F. and Robin, J.-M. (2002). Equilibrium wage dispersion with worker and employer heterogeneity. *Econometrica*, 70(6):2295–2350.
- Rowan, T. M. (1990). The Subplex method for unconstrained optimization. Technical Report ORNL/TM-11504, Oak Ridge National Lab., TN (USA). Cited in NLOpt implementation of Subplex algorithm.
- Rubinstein, A. (1982). Perfect equilibrium in a bargaining model. *Econometrica*, pages 97–109.
- Shi, L. (2023). Optimal regulation of noncompete contracts. *Econometrica*, 91(2):425–463.
- Smith, A. A. (2016). *Indirect Inference*, pages 1–6. Palgrave Macmillan UK, London.
- Taylor, L. A. (2010). Why are CEOs rarely fired? Evidence from structural estimation. *The Journal of Finance*, 65(6):2051–2087.
- Taylor, L. A. (2013). CEO wage dynamics: Estimates from a learning model. *Journal of Financial Economics*, 108(1):79–98.
- Waldman, M. and Zabojnik, J. (2025). Asymmetric learning and the CEO labor market. *Available at SSRN 5551358*.

Table 1. Summary and labor market transition statistics

This table displays summary statistics for managerial wages, and labor market experience and tenure (Panel A); and labor market transition statistics (Panel B). In Panel A, statistics are displayed based on employment status (executive or CEO). In Panel B, labor market transition statistics are displayed conditional on employment state (executive, CEO, unattached); the first two columns display the number of transitions, and the proportion of observations conditional on employment state (i.e., the rate at which the transition occurs cross-sectionally within each group CEO, non-CEO executive, and unattached). The final two columns display the average years of experience and tenure that the transition occurs, respectively.

Panel A: Summary statistics

	N	Mean	Std Dev	25%	50%	75%
CEOs						
Log wage	61,558	14.857	1.208	14.067	14.917	15.685
Wage (\$ million)	61,558	5.342	10.872	1.286	3.007	6.485
Wage Growth (%)	56,245	9.772%	73.908%	-15.123%	7.432%	35.274%
Experience	61,558	22.186	11.719	13	21	31
Tenure	61,558	15.267	11.543	6	12	22
Non-CEO executives						
Log wage	262,162	13.790	1.067	13.051	13.773	14.507
Wage (\$ million)	262,162	1.744	3.905	0.465	0.958	1.997
Wage Growth (%)	210,196	13.734%	64.974%	-11.185%	9.573%	36.749%
Experience	262,162	15.239	11.901	5	12	23
Tenure	262,162	5.776	5.848	2	4	7

Panel B: Labor market transition statistics

	N	%	Experience at transition	Tenure at transition
CEOs				
Horizontal CEO move	293	0.476%	22.631	12.840
Unattachment	543	0.882%	20.722	11.560
Retirement	8,735	14.190%	23.673	15.538
Non-CEO executives				
Internal CEO promotion	4,170	1.591%	17.190	10.474
External CEO promotion	374	0.143%	16.321	5.495
Horizontal executive move	1,535	0.586%	14.383	4.805
Unattachment	5,098	1.945%	15.048	4.595
Retirement	41,169	15.704%	15.680	6.340
Unattached managers				
Executive position	5,085	22.644%	19.197	
CEO position	823	3.665%	22.176	

Table 2. Model fit

This table compares observed and simulated moments, given the data described in Section 2 and the estimation and identification strategies described in Section 4. “Observed” refers to the real data. “Simulated” refers to the simulated data. Standard errors are displayed below each moment in parentheses. Standard errors for the empirical moments are computed using an influence function approach and are clustered at the manager level. Standard errors for the simulated moments are computed using the standard GMM moment variance implied by the simulated data. Moments in Panel A concern labor market mobility and steady-state shares of employment. Moments in Panel B concern the two auxiliary regressions.

Panel A: Labor market mobility

	Description	Observed	Simulated
(1)	Internal CEO hire (0-5 yrs)	0.133 (0.001)	0.135 (0.001)
(2)	Internal CEO hire (5-10 yrs)	0.158 (0.002)	0.119 (0.003)
(3)	External CEO hire (0-5 yrs)	0.020 (0.001)	0.019 (0.000)
(4)	External CEO hire (5-10 yrs)	0.016 (0.001)	0.016 (0.001)
(5)	External exec hire (0-5 yrs)	0.093 (0.001)	0.086 (0.001)
(6)	External exec hire (5-10 yrs)	0.062 (0.002)	0.074 (0.002)
(7)	Unattachment shock (0-5 yrs)	0.066 (0.001)	0.052 (0.001)
(8)	Unattachment shock (5-10 yrs)	0.056 (0.001)	0.061 (0.002)
(9)	Unattachment to exec (0-2 yrs)	0.408 (0.009)	0.416 (0.008)
(10)	Unattachment to CEO (0-2 yrs)	0.068 (0.008)	0.060 (0.017)
(11)	CEO share	0.342 (0.003)	0.345 (0.004)

Panel B: Auxiliary regressions

Description		Observed	Simulated
Auxiliary Mincer regression			
(1)	Tenure (linear)	0.064 (0.005)	0.062 (0.006)
(2)	Tenure (quadratic)	-0.335 (0.036)	-0.290 (0.055)
(3)	Tenure (cubic)	0.514 (0.066)	0.272 (0.113)
(4)	Experience (linear)	0.088 (0.004)	0.112 (0.005)
(5)	Experience (quadratic)	0.019 (0.027)	-0.188 (0.035)
(6)	Experience (cubic)	-0.069 (0.041)	0.282 (0.063)
(7)	Firm effect (mean)	12.411 (0.004)	12.398 (0.004)
(8)	Firm effect (variance)	0.589 (0.005)	0.599 (0.005)
(9)	Firm effect (skewness)	-0.166 (0.023)	-0.235 (0.021)
(10)	Executive effect (variance)	1.551 (0.011)	1.565 (0.011)
(11)	Wage residual (variance)	0.270 (0.004)	0.290 (0.004)
(12)	Wage residual (autocovariance)	0.347 (0.008)	0.335 (0.033)
Auxiliary wage growth regression			
(13)	Tenure growth (quadratic)	-0.371 (0.014)	-0.284 (0.029)
(14)	Tenure growth (cubic)	0.602 (0.032)	0.418 (0.044)
(15)	Experience growth (quadratic)	-0.169 (0.011)	-0.228 (0.019)
(16)	Experience growth (cubic)	0.133 (0.020)	0.263 (0.027)
(17)	Wage growth residual (variance)	0.365 (0.006)	0.392 (0.007)
(18)	Wage growth residual (autocovariance)	-0.377 (0.011)	-0.435 (0.011)

Table 3. Structural parameter estimates

This table displays the estimated structural parameters. Parameter standard errors, derived via the Delta method using the optimal weight matrix, are displayed in parentheses next to each parameter.

Description	Notation	Parameter value
Arrival rates		
Internal CEO offer	λ_0	0.0279 (0.0003)
External CEO offer	λ_1	0.0057 (0.0006)
External exec offer	λ_2	0.0181 (0.0003)
Unattachment shock	η	0.0237 (0.0008)
Unattachment to CEO	γ_C	0.0297 (0.0034)
Unattachment to exec	γ_E	0.2425 (0.0056)
Bargaining Power		
Managerial bargaining power	β	0.2640 (0.0397)
Human capital accumulation		
Tenure polynomial (linear)	δ_k^1	0.1108 (0.0039)
Tenure polynomial (quadratic)	$\delta_k^2 \times 10^2$	-0.3827 (0.0285)
Tenure polynomial (cubic)	$\delta_k^3 \times 10^4$	0.3396 (0.0511)
Experience polynomial (linear)	δ_g^1	0.0570 (0.0046)
Experience polynomial (quadratic)	$\delta_g^2 \times 10^2$	-0.0277 (0.0200)
Experience polynomial (cubic)	$\delta_g^3 \times 10^4$	0.1155 (0.0279)
Firm heterogeneity		
Location	p_{min}	12.2772 (0.0405)
Scale	s_1	2.2217 (0.0695)
Shape	s_2	3.1023 (0.1705)
Manager heterogeneity		
Ability (volatility)	σ_a	1.7610 (0.0150)
Idiosyncratic shock (volatility)	σ_e	0.6676 (0.0026)
Idiosyncratic shock (persistence)	ν_e	0.0720 (0.0255)

Table 4. Dispersion in managerial compensation

We decompose the dynamic components of wage dispersion by using the following identity for wage variance (conditional on permanent manager experience a_i):

$$\begin{aligned} \text{Var}(w_{it} \mid a_i) = & \text{Var}(r_{it} \mid a_i) + \text{Var}(f_{it} \mid a_i) + \text{Var}(g_{it} \mid a_i) + \text{Var}(k_{it} \mid a_i) + \text{Var}(u_{it} \mid a_i) + \\ & 2 \text{Cov}(r_{it}, f_{it} \mid a_i) + 2 \text{Cov}(r_{it}, g_{it} \mid a_i) + 2 \text{Cov}(r_{it}, k_{it} \mid a_i) + \\ & 2 \text{Cov}(f_{it}, g_{it} \mid a_i) + 2 \text{Cov}(f_{it}, k_{it} \mid a_i) + 2 \text{Cov}(g_{it}, k_{it} \mid a_i) \end{aligned}$$

We then display how much of wage dispersion is driven by dispersion in each underlying component. For example, column 1 shows that 34.5% of wage dispersion is driven by piece rates. In columns 2-4 (5-7), we further condition on experience (tenure) bins. Covariance terms with the error u_{it} are dropped as errors are independent of other forces in the model.

	(1) Full sample	(2) Conditional on experience (t)	(3)	(4)	(5) Conditional on tenure (τ)	(6)	(7)
		[0, 5)	[5, 10)	[10, 25)	[0, 5)	[5, 10)	[10, 25)
Var(r)	34.45%	45.16%	58.49%	56.21%	42.26%	43.88%	47.74%
Var(f)	17.28%	37.74%	31.51%	26.26%	20.08%	22.02%	24.82%
Var(g)	10.51%	0.80%	0.75%	6.46%	7.01%	5.84%	6.68%
Var(k)	5.84%	2.35%	2.90%	5.16%	1.29%	0.68%	0.47%
Var(u)	13.82%	29.29%	24.95%	21.47%	15.69%	17.45%	20.47%
2 Cov(r, f)	-5.93%	-17.09%	-10.54%	-9.80%	-6.36%	-6.17%	-10.15%
2 Cov(r, g)	10.47%	1.44%	1.17%	3.68%	19.24%	14.82%	6.89%
2 Cov(r, k)	-0.16%	-1.58%	-10.51%	-13.21%	-0.75%	0.08%	-0.12%
2 Cov(f, g)	1.36%	0.01%	0.19%	0.49%	0.62%	0.51%	0.63%
2 Cov(f, k)	0.89%	-0.29%	-0.16%	0.57%	-0.09%	0.15%	0.15%
2 Cov(g, k)	11.52%	2.50%	1.39%	2.73%	1.14%	1.11%	2.37%

Table 5. Managerial surplus capture decomposition

This table shows realized managerial surplus capture. Panel A displays realized surplus capture by CEO hire type. Panel B displays results across labor market experience and Panel C by firm tenure. In each panel, the first row displays average realized surplus capture b , and the second row displays the share attributable to labor market competition. In Panel A, we focus exclusively on CEOs in the simulated sample, so the averages in column 1 refer only to CEOs. In panels B and C, we focus on the full sample and include executives in the analysis.

Panel A: CEO surplus capture decomposition by CEO hire type

	(1) CEO sample	(2) Conditional on CEO hire type	(3)
		Internal CEO	External CEO
Realized surplus capture (b)	57.91%	49.99%	79.85%
Labor market competition ($\frac{b-\beta}{b}$)	54.41%	47.19%	66.94%

Panel B: Managerial surplus capture decomposition by labor market experience

	(1) Full sample	(2) Conditional on experience (t)	(3)	(4)
		[0, 5)	[5, 10)	[10, 25)
Realized surplus capture (b)	49.47%	40.74%	46.50%	53.46%
Labor market competition ($\frac{b-\beta}{b}$)	46.64%	35.20%	43.23%	50.62%

Panel C: Managerial surplus capture decomposition by firm tenure

	(1) Full sample	(2) Conditional on tenure (τ)	(3)	(4)
		[0, 5)	[5, 10)	[10, 25)
Realized surplus capture (b)	49.47%	47.01%	48.05%	51.95%
Labor market competition ($\frac{b-\beta}{b}$)	46.64%	43.85%	45.06%	49.18%

A. Model Appendix

Proof of Theorem 1. This proof is essentially a restatement of the proof of Proposition 2 in [Cahuc et al. \(2006\)](#). Consider a manager in a position of type $i \in \{E, C\}$ who is approached by an outside firm to serve in another type $j \in \{E, C\}$ position. Upon the arrival of the offer, the poacher, incumbent, and manager initiate a bargaining game with the following structure:

1. Stage 1: Both firms simultaneously offer a piece rate to the executive.
2. Stage 2: The executive chooses one of the offers, or rejects and keeps their current position.
3. Stage 3: If the executive accepted an offer in Stage 2, some time elapses. The executive then renegotiates with the firm whose offer was rejected, where the renegotiation protocol mirrors that of the internal bargaining protocol detailed in the Internet Appendix. Unlike with internal negotiations, however, the executive's outside option is not their current position, but the offer accepted in Stage 2.

The bargaining game is solved via backward induction. Let r'_1 and r_1 denote the respective Stage 1 offers from the poacher and incumbent. Suppose that r_1 (the incumbent's initial offer) was accepted in the second stage, triggering a Stage 3 renegotiation with the poacher. The counteroffer will satisfy:

$$V_j(r, t, f') = \beta V_j(0, t, f') + (1 - \beta) V_i(r_1, t, p) \quad (\text{A.1})$$

Conversely, suppose that r'_1 was accepted in Stage 2. In the subsequent renegotiation with the incumbent firm, their counteroffer will satisfy:

$$V_i(r, t, p) = \beta V_i(0, t, p) + (1 - \beta) V_j(r'_1, t, f') \quad (\text{A.2})$$

The form of counteroffers implies that if r_1 was accepted in Stage 2, the manager optimally renegotiates and eventually works with the poacher if and only if $V_j(0, t, f') > V_i(r_1, t, p)$. Alternatively, if r'_1 was accepted in Stage 2, the manager renegotiates and eventually works with the incum-

bent if and only if $V_i(0, t, p) \geq V_j(r'_1, t, f')$. Thus, the values of accepting r_1 and r'_1 at Stage 2 are respectively given by:

$$\max\{\beta V_j(0, t, f') + (1 - \beta)V_i(r_1, t, p), V_i(r_1, t, p)\} \quad (\text{A.3})$$

$$\max\{\beta V_i(0, t, p) + (1 - \beta)V_i(r'_1, t, f'), V_j(r'_1, t, f')\} \quad (\text{A.4})$$

Moving back to Stage 1, both firms make simultaneous offers. Noting that firm profits are non-negative if and only if $r \leq 0$, for the poacher to eventually win the manager, they must bid r'_1 such that $V_j(r'_1, t, f') > V_i(0, t, p)$, so the incumbent cannot afford to outbid. Suppose momentarily that $i = j$. If $f' > p$, were the manager to accept $r_1 = 0$ from the incumbent in stage 1, they would eventually renegotiate with the poacher in stage 3 following:

$$V_j(r'_1, t, f') = \beta V_j(0, t, f') + (1 - \beta)V_i(0, t, p) \quad (\text{A.5})$$

To deter subsequent renegotiation with the incumbent, a poacher with $f' > p$ immediately bids r'_1 satisfying (A.5), which the manager rationally accepts. Thus, for $i = j$ (lateral moves), equilibrium bidding behavior implies that all outside offers such that $f' > p$ are accepted. For $i = E$ and $j = C$, we simply substitute $V_i(0, t, p)$ for $V_j(0, t, \bar{\theta}(0, p))$ in Equation (A.5) which implies all external promotions such that $f' > \bar{\theta}(0, p)$ are accepted.

Things are symmetric if $f' \leq p$ ($f' \leq \bar{\theta}(0, p)$) in case of lateral moves (external promotions). Focusing on lateral moves, if $f' \leq p$, the incumbent immediately bids r_1 such that:

$$V_i(r_1, t, p) = \beta V_i(0, t, p) + (1 - \beta)V_i(0, t, f') \quad (\text{A.6})$$

which similarly deters subsequent renegotiation with the poacher, and the manager rationally accepts. Note additionally that a competing offer does not necessitate an alteration of the current

piece rate r . The minimal value of f' such that “something happens” is defined by:

$$V_i(r, t, p) = \beta V_i(0, t, p) + (1 - \beta) V_j(0, t, \underline{\theta}_{ij}(r, p)) \quad (\text{A.7})$$

■

A.1. Value Function Derivations

Given the acceptance criteria summarized in Theorem 1, we next characterize the value functions associated with CEO and executive employment. We assume that managers’ flow utility is logarithmic in their wage. We also impose the following transversality condition:

$$\lim_{p \rightarrow \infty} \frac{\partial V_C}{\partial p}(r, t, p) = \frac{1}{\rho + \mu + \eta} = \frac{1}{\gamma} \quad (\text{A.8})$$

where we define $\gamma = \rho + \mu + \eta$ for brevity.

A.1.1. CEO Value Function

Let $S(\cdot) = 1 - \Psi(\cdot)$ be the survivor function for the distribution of firm productivity. The value of CEO employment satisfies:

$$\begin{aligned} (\gamma + \lambda_1 S(\underline{\theta}_{CC})) V_C(r, t, p) &= w + \frac{\partial V_C}{\partial t}(r, t, p) + \delta_k \frac{\partial V_C}{\partial p}(r, t, p) + \eta V_U(t) + \\ &\lambda_1 \int_p^\infty \left[(1 - \beta) V_C(0, t, p) + \beta V_C(0, t, x) \right] dF(x) + \lambda_1 \int_{\underline{\theta}_{CC}}^p \left[(1 - \beta) V_C(0, t, x) + \beta V_C(0, t, p) \right] dF(x) \end{aligned} \quad (\text{A.9})$$

Rearranging (A.9) via integration by parts yields:

$$\begin{aligned} \gamma V_C(r, t, p) &= w + \eta V_U(t) + \frac{\partial V_C}{\partial t}(r, t, p) + \delta_k \frac{\partial V_C}{\partial p}(r, t, p) + \lambda_1 \beta \int_p^\infty \frac{\partial V_C}{\partial x}(0, t, x) S(x) dx \\ &\quad + \lambda_1 (1 - \beta) \int_{\underline{\theta}_{CC}}^p \frac{\partial V_C}{\partial x}(0, t, x) S(x) dx \end{aligned} \quad (\text{A.10})$$

Consequently, the value function evaluated at $r = 0$ is given by:

$$\gamma V_C(0, t, p) = g + f + k + \eta V_U(t) + \frac{\partial V_C}{\partial t}(0, t, p) + \delta_k \frac{\partial V_C}{\partial p}(0, t, p) + \lambda_1 \beta \int_p^\infty \frac{\partial V_C}{\partial x}(0, t, x) S(x) dx \quad (\text{A.11})$$

Differentiating with respect to p yields:

$$(\gamma + \lambda_1 \beta S(p)) \frac{\partial V_C}{\partial p}(0, t, p) = 1 + \delta_k \frac{\partial^2 V_C}{\partial p^2}(0, t, p) \quad (\text{A.12})$$

Note that the cross partial $\frac{\partial^2 V_C}{\partial p \partial t}(r, t, p)$ vanishes as the effects of t and p are separable. Next, define $\phi_C(x) \equiv \frac{\partial V_C}{\partial p}(0, t, x)$. Additionally, let $\omega_C(x) = \gamma + \lambda_1 \beta S(p)$. Then Equation (A.12) can be written as:

$$\omega_C(x) \phi_C(x) = 1 + \delta_k \phi'_C(x) \quad (\text{A.13})$$

This is a first-order linear ODE in x which can be solved explicitly. Define $\Omega_C(x) = \exp\left(-\frac{1}{\delta_k} \int_{f_{\min}}^x \omega_C(z) dz\right)$.

Rearranging (A.13), multiplying both sides by $\Omega_C(x)$, and solving yields:

$$\phi_C(p) = \frac{1}{\Omega_C(p)} \left[C - \frac{1}{\delta_k} \int_{f_{\min}}^p \Omega_C(x) dx \right] \quad (\text{A.14})$$

The constant of integration C is pinned down by the transversality condition (A.8). Letting $C = \frac{1}{\delta_k} \int_{f_{\min}}^\infty \Omega_C(x) dx$,

we can use L'Hospital's rule to verify:

$$\lim_{p \rightarrow \infty} \phi_C(p) = \lim_{p \rightarrow \infty} \frac{\frac{1}{\delta_k} \int_p^\infty \Omega_C(x) dx}{\Omega_C(p)} = \lim_{p \rightarrow \infty} \frac{1}{\omega_C(p)} = \frac{1}{\gamma} \quad (\text{A.15})$$

A.1.2. Executive Value Function

The procedure for the executive value function is the same as before, though notation is slightly more cumbersome. The net present value of executive employment can be represented as:

$$\begin{aligned} (\gamma + \lambda_0 + \lambda_1 S(\underline{\theta}_{EC}) + \lambda_2 S(\underline{\theta}_{EE})) V_E(r, t, p) &= w + \frac{\partial V_E}{\partial t}(r, t, p) + \delta_k \frac{\partial V_E}{\partial p}(r, t, p) + \eta V_U(t) + \\ \lambda_0 \left[(1 - \beta) V_E(r, t, p) + \beta V_C(0, t, p) \right] &+ \lambda_1 \int_{\hat{\theta}(p)}^\infty \left[(1 - \beta) V_E(0, t, p) + \beta V_C(0, t, x) \right] dF(x) + \\ \lambda_2 \int_{\underline{\theta}_{EE}}^p \left[(1 - \beta) V_E(0, t, x) + \beta V_E(0, t, p) \right] &dF(x) \end{aligned} \quad (\text{A.16})$$

Noting that $V_C(r, t, f_{\min}) = V_U(t)$ by assumption, we can (as in the CEO case) rearrange (A.16) via integration by parts:

$$\begin{aligned}
(\gamma + \lambda_0\beta)V_E(r, t, p) &= w + \frac{\partial V_E}{\partial t}(r, t, p) + \delta_k \frac{\partial V_E}{\partial p}(r, t, p) + (\eta + \lambda_0\beta)V_U(t) + \lambda_0\beta \int_{f_{\min}}^p \phi_C(x)dx \\
&\quad + \lambda_1\beta \int_{\bar{\theta}(0, p)}^{\infty} \phi_C(x)S(x)dx + \lambda_1(1 - \beta) \int_{\underline{\theta}_{EC}}^{\bar{\theta}(0, p)} \phi_C(x)S(x)dx \\
&\quad + \lambda_2\beta \int_p^{\infty} \frac{\partial V_E}{\partial x}(0, t, x)S(x)dx + \lambda_2(1 - \beta) \int_{\underline{\theta}_{EE}}^p \frac{\partial V_E}{\partial x}(0, t, x)S(x)dx
\end{aligned} \tag{A.17}$$

Again setting $r = 0$ and differentiating with respect to p yields:

$$\frac{\partial V_E}{\partial p}(0, t, p) = \frac{1 + \delta_k \frac{\partial^2 V_E}{\partial p^2}(0, t, p) + \lambda_0\beta\phi_C(p)}{\delta + \lambda_0\beta + \lambda_1\beta S(\bar{\theta}(p)) + \lambda_2\beta S(p)} \tag{A.18}$$

Similar to before, defining $\phi_E(p) \equiv \frac{\partial V_E}{\partial p}(0, t, p)$ leaves us with a first-order linear ODE in p whose solution is given by:

$$\phi_E(p) = \frac{1}{\Omega_E(p)} \left[C - \frac{1}{\delta_k} \int_{p_{\min}}^p \Omega_E(x) \left(1 + \lambda_0\beta\phi_C(x) \right) dx \right] \tag{A.19}$$

where $\omega_E(p) = \gamma + \lambda_0\beta + \lambda_1\beta S(\bar{\theta}(0, p)) + \lambda_2\beta S(p)$ and $\Omega_E(p) = \exp\left(-\frac{1}{\delta_k} \int_{f_{\min}}^p \omega_E(x)dx\right)$. Similar to the case for CEOs, letting $C = \frac{1}{\delta_k} \int_{f_{\min}}^{\infty} \Omega_E(x)(1 + \lambda_0\beta\phi_C(x))dx$, we have that:

$$\lim_{p \rightarrow \infty} \phi_E(p) = \lim_{p \rightarrow \infty} \frac{\frac{1}{\delta_k} \int_p^{\infty} \Omega_E(x)(1 + \lambda_0\beta\phi_C(x))dx}{\Omega_E(p)} = \lim_{p \rightarrow \infty} \frac{1 + \lambda_0\beta\phi_C(p)}{\omega_E(p)} = \frac{1 + \lambda_0\beta\gamma^{-1}}{\gamma + \lambda_0\beta} \tag{A.20}$$

Lemma 1. $\phi_C(p) > \phi_E(p)$

Proof. Define $d(p) := \phi_C(p) - \phi_E(p)$. Subtracting (A.12) from (A.18) and writing $\omega_E(p) = \omega_C(p) + \Delta(p)$, $\Delta(p) > 0$, yields:

$$\begin{aligned}
\delta_k d'(p) &= \omega_C(p) d(p) - \Delta(p) \phi_E(p) - \lambda_0\beta \phi_C(p). \\
d'(p) - \frac{\omega_C(p)}{d(p)} &= -\frac{1}{\delta_k} [\Delta(p) \phi_E(p) + \lambda_0\beta \phi_C(p)]
\end{aligned} \tag{A.21}$$

Because every term in the bracket is non-negative and at least one is strictly positive, the right-hand side

of (A.21) is smaller than $\omega_C(p)d(p)$ whenever $d(p) \leq 0$. Multiply (A.21) by $\Omega_C(p)$ to obtain:

$$\frac{d}{dp} [\Omega_C(p) d(p)] = -\frac{\Omega_C(p)}{\delta_k} [\Delta(p) \phi_E(p) + \lambda_0 \beta \phi_C(p)] < 0 \quad (\text{A.22})$$

Integrating from p to ∞ and using $\Omega_C(\infty)d(\infty) = 0$ yields:

$$d(p) \Omega_C(p) = \int_p^\infty \frac{\Omega_C(x)}{\delta_k} [\Delta(x) \phi_E(x) + \lambda_0 \beta \phi_C(x)] dx > 0 \quad (\text{A.23})$$

Since $\Omega_C(p) > 0$, $d(p) = \phi_C(p) - \phi_E(p) > 0$ for all $p < \infty$. ■

Proof of Proposition 1. By the previous lemma, we know that $\phi_E(x) < \phi_C(x)$ for all $x \in [f_{\min}, \infty)$. Next, define $D(p) = \bar{\theta}(0, p) - p$. Given that $\bar{\theta}(0, f_{\min}) = f_{\min}$, we can prove that $\bar{\theta}(0, p) < p$ by establishing that $D(p) \leq 0$ for all $p \in [f_{\min}, \infty)$. For the sake of contradiction, suppose there exists a p^* such that $D(p^*) > 0$. As $D(p)$ is continuous and $D(f_{\min}) = 0$, there must in this case be a point p_0 such that $D(p_0) = 0$ and $D(p) > 0$ for all $p \in (p_0, p^*]$. Further, it must be true that $D'(p_0) \geq 0$. From the definition of $D(p)$, we know that:

$$D'(p_0) = \frac{\phi_E(p_0)}{\phi_C(\bar{\theta}(0, p_0))} - 1 \quad (\text{A.24})$$

Given that $\phi_C(x)$ is monotonic, if $D(p_0) = 0$, then we must have that $\bar{\theta}(0, p_0) = p_0$. This then implies that:

$$D'(p_0) = \frac{\phi_E(p_0)}{\phi_C(p_0)} - 1 < 0 \quad (\text{A.25})$$

where the inequality follows from the fact that $\phi_E(x) < \phi_C(x)$ (Lemma 1) for all $x \in [f_{\min}, \infty)$. This contradicts the original assumption that $D'(p_0) \geq 0$. Hence, it must be true that $D(p) \leq 0$ for all $p \in [f_{\min}, \infty)$. Finally, to establish that $\bar{\theta}(r, p) < \bar{\theta}(0, p)$, note that by the fundamental theorem of calculus:

$$\bar{\theta}(r, p) = \bar{\theta}(0, p) - \int_{-\infty}^r \frac{\partial \bar{\theta}}{\partial s}(s, p) ds \quad (\text{A.26})$$

Differentiating the indifference condition (5) with respect to r yields:

$$\frac{\partial \bar{\theta}}{\partial r}(r, p) = \frac{\frac{\partial V_E}{\partial r}(r, t, p)}{\phi_C(\bar{\theta}(r, p))} \quad (\text{A.27})$$

Rewriting Equation (A.26):

$$\bar{\theta}(r, p) = \bar{\theta}(0, p) - \int_{-\infty}^r \frac{\frac{\partial V_E}{\partial s}(s, t, p)}{\phi_C(\bar{\theta}(s, p))} ds \quad (\text{A.28})$$

As both $\frac{\partial V_E}{\partial s}(s, t, p) > 0$ and $\phi_C(\bar{\theta}(s, p)) > 0$, it must be the case that $\bar{\theta}(r, p) < \bar{\theta}(0, p)$ for $r < 0$. ■

Proof of Proposition 2. The equilibrium piece rates follow immediately from substituting the value functions above into their associated bargaining conditions and rearranging terms. We focus on the case of lateral CEO moves in the interest of brevity. The bargaining condition (10) implies:

$$\begin{aligned} r' = & -(1 - \beta)(f' - p) + \beta \left[\frac{\partial V_C}{\partial t}(0, t, f') + \delta_k \frac{\partial V_C}{\partial p}(0, t, f') \right] \\ & + (1 - \beta) \left[\frac{\partial V_C}{\partial t}(0, t, p) + \delta_k \frac{\partial V_C}{\partial p}(0, t, p) \right] - \frac{\partial V_C}{\partial t}(r', t, f') - \delta_k \frac{\partial V_C}{\partial p}(r', t, f') \\ & + \lambda_1 \beta^2 \int_{f'}^{\infty} \phi_C(x) S(x) dx + \lambda_1 \beta (1 - \beta) \int_{f'}^{\infty} \phi_C(x) S(x) dx \\ & - \lambda_1 \beta \int_{f'}^{\infty} \phi_C(x) S(x) dx - \lambda_1 (1 - \beta) \int_p^{f'} \phi_C(x) S(x) dx \end{aligned} \quad (\text{A.29})$$

By the Envelope Theorem, (10) implies the following two conditions:

$$\frac{\partial V_C}{\partial t}(r', t, f') = \beta \frac{\partial V_C}{\partial t}(0, t, f') + (1 - \beta) \frac{\partial V_C}{\partial t}(0, t, p) \quad (\text{A.30})$$

$$\delta_k \frac{\partial V_C}{\partial p}(r', t, f') = \delta_k \beta \frac{\partial V_C}{\partial p}(0, t, f') + \delta_k (1 - \beta) \frac{\partial V_C}{\partial p}(0, t, p) \quad (\text{A.31})$$

The partial derivative terms in Equation (A.29) then cancel, and we can simplify the expression by combining integrals:

$$r' = -(1 - \beta)(f' - p) - \lambda_1 (1 - \beta)^2 \int_p^{f'} \phi_C(x) S(x) dx = -(1 - \beta) \int_p^{f'} q(x) dx \quad (\text{A.32})$$

$$q(x) = 1 + \lambda_1 (1 - \beta) \phi_C(x) S(x) \quad (\text{A.33})$$

The procedure for the remaining piece rates is identical. ■

Proof of Proposition 3. This proof is identical to the Proof of Proposition 2, except we substitute the value functions into the renegotiation condition (11) as opposed to (10). ■

Proof of Proposition 4. We begin by characterizing the dynamics of θ_{CC} . We restrict attention to the case

of lateral CEO transitions as the procedure for the other cases is identical. Recall the definition of $q(x)$ in Equation (A.33). Differentiating $q(x)$ yields:

$$q'(x) = \lambda_1(1 - \beta)[\phi'_C(x)S(x) - \phi_C(x)\psi(x)] \quad (\text{A.34})$$

where $\psi(x)$ is the PDF corresponding to $\Psi(x)$. By the definition of $\phi_C(x)$ (Equation (A.14)), we have that $\phi'_C(x) \leq 0$ (i.e. value function is weakly concave in p). By Equation (A.34), this then implies that $q(x)$ is weakly decreasing. Next, note that Equation (9) implies the identity:

$$\int_z^f q(x)dx = \int_{\theta_{CC}(r,p)}^p q(x)dx \quad (\text{A.35})$$

Differentiating both sides with respect to p yields:

$$\frac{\partial \theta_{CC}}{\partial p}(r, p) = \frac{q(p)}{q(\theta_{CC}(r, p))} > 0 \quad (\text{A.36})$$

Noting that $\frac{\partial \theta_{CC}}{\partial \tau}(r, p) = \frac{\partial \theta_{CC}}{\partial p}(r, p) \frac{dk}{d\tau}(\tau)$, we have that:

$$\frac{\partial \theta_{CC}}{\partial \tau}(r, p) = \frac{dk}{d\tau}(\tau) \frac{q(p)}{q(\theta_{CC}(r, p))} > 0 \quad (\text{A.37})$$

As the upper bound p of the renegotiation set $[\theta_{CC}(r, p), p]$ grows at rate $\frac{dk}{d\tau}(\tau)$, whether the mass of the interval $[\theta_{CC}(r, p), p]$ increases or decreases with tenure depends on the magnitude of the ratio $\frac{q(p)}{q(\theta_{CC}(r, p))}$.

As $q(x)$ is weakly decreasing, it follows that $\frac{q(p)}{q(\theta_{CC}(r, p))} \leq 1$. Therefore, we have that $\frac{\partial \theta_{CC}}{\partial \tau} \leq \frac{dk}{d\tau}(\tau) = \frac{\partial x_{CC}}{\partial \tau}$.

The procedure is identical for the other cases $\{EC, EE\}$. ■

B. Appendix Figures and Tables

Figure B.1. Firm heterogeneity distribution

This figure displays the estimated position productivity distribution, which is characterized by location f_{min} , shape s_1 and scale s_2 from Table 3. In the figure, we display the (simulated) empirical CDFs of position-types, conditional on experience t .

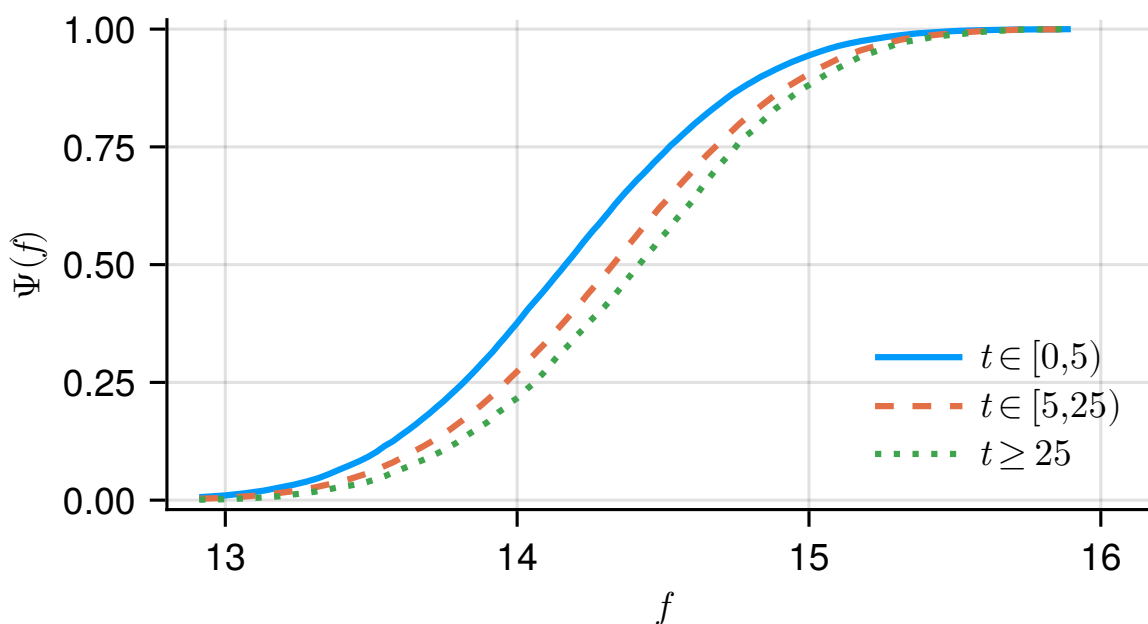


Figure B.2. Firm-specific human capital and transition rates

This figure displays cumulative hazard rates of a manager experiencing (A) an external CEO move and (B) internal promotion. The blue line displays the estimated rates, the orange lines the counterfactual when $k = 0$. For example, roughly .5% (1%) of managers are poached into a CEO position within 10 years of tenure in the baseline (counterfactual).

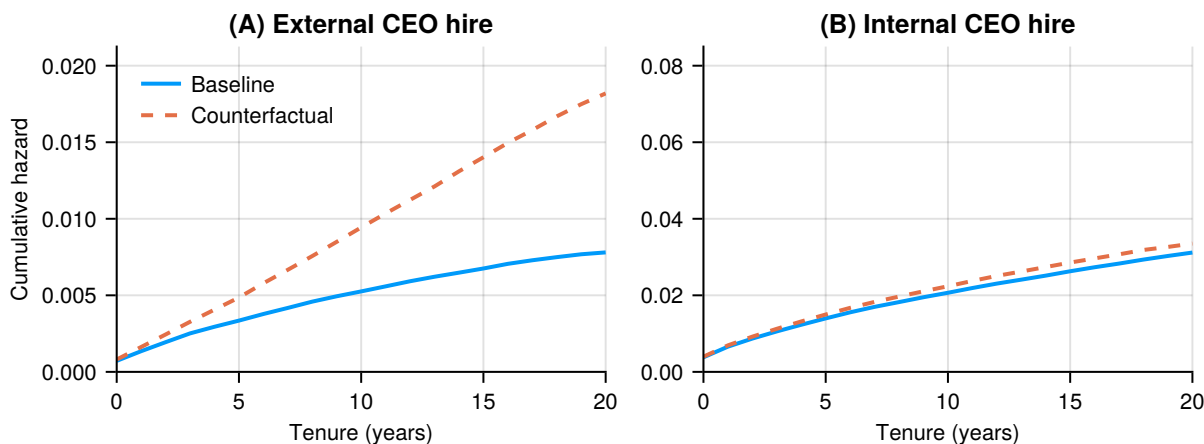


Table B.1. Managerial human capital composition

This table shows a decomposition of accumulated CEO human capital into general human capital $g(t)$, and firm-specific human capital $k(\tau)$ across labor market experience t and across tenure at a firm τ . We compute a variance decomposition: $s_x = \text{Cov}(h, x) / \text{Var}(h)$ for $x \in \{g, k\}$.

	(1)	(2)	(3)	(4)
	Full sample	Conditional on experience (t)		
		[0, 5)	[5, 10)	[10, 25)
General human capital (g)	55.08%	34.95%	24.94%	59.20%
Firm-specific human capital (k)	44.92%	65.05%	75.06%	40.80%
		Conditional on tenure (τ)		
General human capital (g)		62.29%	75.09%	81.67%
Firm-specific human capital (k)		37.71%	24.91%	18.33%

Internet Appendix

IA1. Estimation Appendix

IA1.1. Weighting Matrix

From the empirical sample we obtain a $K \times 1$ vector of moments $\hat{M} \in \mathbb{R}^K$. Let $\psi_n \in \mathbb{R}^K$ denote the influence function of observation n on those moments. Observations are grouped by firm; let $f \in \{1, \dots, F\}$ indicate firms and $n \in f$ for the N_f observations which belong to firm g . We first stack the within-firm influence functions:

$$\Psi_f = \sum_{n \in f} \psi_n \in \mathbb{R}^K \quad (\text{IA1})$$

$$\Psi = [\Psi_1, \Psi_2, \dots, \Psi_F]^T \quad (\text{IA2})$$

The clustered covariance matrix of the vector of moments can then be estimated as:

$$\widehat{avar}(\hat{M}) = \frac{1}{F} \sum_{f=1}^F \Psi_f \Psi_f^T \quad (\text{IA3})$$

The weighting matrix $\hat{W}$ is then obtained as the inverse of matrix [IA3](#). Let $\Theta \in \mathbb{R}^P$ denote an arbitrary vector of structural parameters. Define the moment residual $g : \mathbb{R}^P \rightarrow \mathbb{R}^M$ as:

$$g(\Theta) = \hat{M} - \frac{1}{S} \sum_{s=1}^S \hat{m}^s(\Theta) \quad (\text{IA4})$$

Where $\hat{M}$ is the vector of empirical moments, $\hat{m}^s(\Theta)$ is the vector of simulated moments given parameter values Θ in simulation s , and S is the total number of simulations. The vector of estimates $\hat{\Theta}$ minimizes the SMM objective function:

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta) \hat{W} g(\Theta)' \quad (\text{IA5})$$

IA1.2. Model Estimation Algorithm

We minimize the SMM quadratic form using the TikTak algorithm. The routine proceeds as follows:

1. *Set initial guesses for model parameters:* We set bounds for each parameter and draw a sequence of 100,000 Sobol points from the resulting bounded parameter space. We then simulate the model, construct our simulated panel, and compute the SMM objective at each Sobol point.
2. *Parallel local minimization:* We then rank the Sobol points by objective function value, and compute the local minimum around the top 225 points (225 is motivated by core usage on the cluster we use). We compute local minima using the [Subplex algorithm via NLOpt](#) (Rowan, 1990). We use 500 iterations for each of the 225 local minima, as we do not need to find the exact local minimum, rather just be in the neighborhood. We rank these 225 local minima by objective function value.
3. *Parallel global minimization:* We then compute convex combinations of each local minima with the current best point: for $i \dots N_S, i > 1$, we set $\theta_2 = 0.995$ and $\theta_{N_S} = 0.005$ as the maximum and minimum weight on the current best local minimum and the remaining Sobol points (ranked 2 to 225). We then compute the local minimum of each candidate global minimum point, using 1000 iterations of the Subplex algorithm. If $i^* > 1$, we update to the position of the best possible global minimum point and repeat the parallel global minimization by re-computing the convex combination of the current best candidate and all remaining points. We do this until i^* is stable between iterations

IA1.3. Standard Errors for Parameter Estimates

For true parameter vector Θ and consistent estimate $\hat{\Theta}$, we have the following asymptotic distribution (Duffie and Singleton, 1993):

$$\sqrt{n}(\hat{\Theta} - \Theta) \rightarrow^d N(0, \text{avar}(\hat{\Theta})) \quad (\text{IA6})$$

$\text{avar}(\hat{\Theta})$ can be expressed as:

$$\text{avar}(\hat{\Theta}) = \left(1 + \frac{1}{S}\right) \left(\frac{\partial g(\Theta)}{\partial \Theta} W \frac{\partial g(\Theta)}{\partial \Theta'} \right)^{-1} \quad (\text{IA7})$$

where $\frac{\partial g(\Theta)}{\partial \Theta}$ is the Jacobian of the moment residual (IA4) with respect to the structural parameters, W is the optimal weighting matrix, and S is the number of simulations. We approximate the Jacobian using:

$$\frac{\partial \hat{g}_m(\Theta)}{\partial \Theta_p} = \frac{g_p(\hat{\Theta} + h_p) - g_p(\hat{\Theta})}{h_p} \quad (\text{IA8})$$

for each moment m and parameter p . h_p is the perturbation size for parameter which we set to 1% of the absolute value of the parameter estimate. The standard errors are the square root of the diagonal elements of the matrix:

$$\left(1 + \frac{1}{S}\right) \left(\frac{\partial \hat{g}(\Theta)}{\partial \Theta} \hat{W} \frac{\partial \hat{g}(\Theta)}{\partial \Theta'} \right)^{-1} \quad (\text{IA9})$$

where $\hat{W}$ is the sample counterpart of the optimal weighting matrix.

IA2. Additional Model Appendix

IA2.1. Derivation of Bargaining Rules

Derivation of sharing rule for internal promotions. Suppose an executive is approached by their firm and offered a promotion to CEO. Each party makes alternating offers over the piece rate r' . If the offer is accepted, the bargaining game ends. If the offer is rejected, some time elapses

before a counteroffer is made. Let $\Delta_e = \beta\epsilon$ and $\Delta_f = (1 - \beta)\epsilon$ respectively denote the lengths of time which elapse following a rejection by the executive and firm. It is also assumed that during negotiations, the match severs at rate ξ in which case the executive remains in their current position, and additional offers for outside CEO and non-CEO positions respectively arrive at rates λ_1 and λ_2 . The subgame perfect equilibrium of this game consists of piece rate offers (r_e, r_f) which make the other party indifferent between immediate acceptance and waiting to make a counteroffer. That is, r_e and r_f respectively solve:

$$V_C(r_f, t, p) = \frac{1}{1 + \rho\Delta_e} \left[w_t\Delta_e + \xi\Delta_e V_E(r, t + \Delta_e, p + \Delta_e k') + \lambda_1\Delta_e \tilde{V}_C(\cdot) + \lambda_2\Delta_e \tilde{V}_E(\cdot) \right. \\ \left. + (1 - \Delta_e(\xi + \lambda_1 + \lambda_2))V_C(r_e, t + \Delta_e, p + \Delta_e k') \right] \quad (\text{IA1})$$

$$\Pi_C(r_e, t, p) = \frac{1}{1 + \rho\Delta_f} \left[\pi_0\Delta_f + \xi\Delta_f\Pi_0 + \lambda_0\Delta_f\tilde{\Pi}_C(\cdot) + \lambda_1\Delta_f\tilde{\Pi}_C(\cdot) \right. \\ \left. + (1 - \Delta_f(\xi + \lambda_0 + \lambda_1))\Pi_C(r_f, t + \Delta_f, p + \Delta_f k') \right] \quad (\text{IA2})$$

$\Pi_C(x)$ denotes the value to the firm of filling the CEO position given state x . π_0 and Π_0 denote the flow and net present values to the firm of having a vacant CEO position, both of which we assume to equal 0. $\tilde{V}_C$ and $\tilde{V}_E$ denote the executive's net present value of initiating a new bargaining game for a CEO or non-CEO position upon the arrival of a competing offer. Similar for $\tilde{\Pi}_C$. The two equations above can be rewritten as:

$$V_C(r_f, t, p) - V_C(r_e, t + \Delta_e, p + \Delta_e k') = -\Delta_e \left[(\xi + \lambda_1 + \lambda_2)V_C(r_e, t + \Delta_e, p + \Delta_e k') + \rho V_C(r_f, t, p) \right. \\ \left. - w_t - \xi V_E(r, t + \Delta_e, p + \Delta_e k') - \lambda_1 \tilde{V}_C(\cdot) - \lambda_2 \tilde{V}_E(\cdot) \right] \quad (\text{IA3})$$

$$\Pi_C(r_e, t, p) - \Pi_C(r_f, t + \Delta_f, p + \Delta_f k') = -\Delta_f \left[(\xi + \lambda_0 + \lambda_1)\Pi_C(r_f, t + \Delta_f, p + \Delta_f k') + \rho \Pi_C(r_e, t, p) \right. \\ \left. - \pi_0 - \xi \Pi_0 - \lambda_0 \tilde{\Pi}_C(\cdot) - \lambda_1 \tilde{\Pi}_E(\cdot) \right] \quad (\text{IA4})$$

The above conditions imply that $r_f \rightarrow r_e$ as $\epsilon \rightarrow 0$. Denote their common limit by r' and

define:

$$\frac{\partial V_C}{\partial r}(r, t, p) = \lim_{\epsilon \rightarrow 0} \frac{V_C(r_f, t, p) - V_C(r_e, t + \Delta_e, p + \Delta_e k')}{r_f - r_e} \quad (\text{IA5})$$

$$\frac{\partial \Pi_C}{\partial r}(r, t, p) = \lim_{\epsilon \rightarrow 0} \frac{V_C(r_f, t, p) - V_C(r_e, t + \Delta_f, p + \Delta_f k')}{r_f - r_e} \quad (\text{IA6})$$

Using the definitions above and taking the ratios of (IA3) and (IA4) yields:

$$-\frac{\frac{\partial V_C}{\partial r}(r', t, p)}{\frac{\partial \Pi_C}{\partial r}(r', t, p)} = \frac{\Delta_e(\rho + \xi + \lambda_1 + \lambda_2)}{\Delta_f(\rho + \xi + \lambda_0 + \lambda_1)} \frac{V_C(r', t, p) - \frac{w_t + \xi V_E(r, t + \Delta_e, p + \Delta_e k') + \lambda_1 \tilde{V}_C(\cdot) + \lambda_2 \tilde{V}_E(\cdot)}{\rho + \xi + \lambda_1 + \lambda_2}}{\Pi_C(r', t, p) - \frac{\pi_0 + \xi \Pi_0 + \lambda_0 \tilde{\Pi}_C(\cdot) + \lambda_1 \tilde{\Pi}_E(\cdot)}{\rho + \xi + \lambda_0 + \lambda_1}} \quad (\text{IA7})$$

Next, define $S(t, p) = \Pi_C(r', t, p) + V_C(r', t, p) - V_E(r, t, p)$ as the surplus associated with the position. Note that $\Pi_C(0, t, p) = 0$, which implies that $\Pi_C(r', t, p) = V_C(0, t, p) - V_C(r', t, p)$. Thus, $\frac{\partial \Pi_C}{\partial r}(r', t, p) = -\frac{\partial V_C}{\partial r}(r', t, p)$. Applying this to (IA7) and taking the limit as $\xi \rightarrow \infty$ yields (after some algebra):

$$V_C(r', t, p) = \beta V_C(0, t, p) + (1 - \beta) V_E(r, t + \Delta_e, p + \Delta_e k') \quad (\text{IA8})$$

Finally, taking $\epsilon \rightarrow 0$ yields the continuous time limit:

$$V_C(r', t, p) = \beta V_C(0, t, p) + (1 - \beta) V_E(r, t, p) \quad (\text{IA9})$$